

Exact Relation Lattices, CM Class Products, and Zero Transitions

Further Results on Polylogarithmic Arithmetic Bridges

Research continuation prepared for Vladimir Reshetnikov

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Abstract

We continue the ProveIt polylogarithms manuscript at the explicitly pinned commit `9bc738d3be22b8586a24693f19fb2e2a50ecd1bf`. The odd-weight rank conjecture for its level-four imaginary depth-two product matrix is proved for every odd weight and strengthened to an integral Smith-form theorem covering both parities. In even weight the same matrices have full column rank and only elementary 2-torsion; a concrete mixed-color evaluation is obtained at every even weight. A resultant formula, derived with explicit Chowla–Selberg normalization, proves the five recorded CM weight-four class products and evaluates their analogues at every even weight. A genus-character refinement proves the three recorded class ratios and gives explicit weight-six ratio identities. For a fixed admissible genus character, all even-weight ratios lie in one number field of degree at most twelve over $\mathbb{Q}$; the three worked examples have quadratic closure. On the analytic side, we prove that the positive zero counts of $\gamma_5^{(k)}$ are $3, 3, 5, 5, \dots$, with every zero simple. Exact rational certificates establish the finite signs needed for this global statement. We also classify all positive zeros near the elementary Lerch endpoint, prove alternating directions of motion for odd-index first derivatives, and establish a nondegenerate creation of two real zeros at index three. Two specific textual corrections, an explicit distinction between proofs and numerical evidence, and a program of further research complete the article. Companion source code and machine-readable receipts make the finite algebraic and analytic claims independently reproducible.

Keywords. Colored multiple polylogarithms; double shuffle; Smith normal form; complex multiplication; gamma products; generalized Stieltjes constants; Lerch transcendent; certified zero isolation.

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1 Scope, conventions, and the status of the results

The point of departure is the 228-page collective manuscript in `Analysis/Polylogarithms/docs/manuscript` of the ProveIt repository [1]. The source was pinned before starting this continuation:

9bc738d3be22b8586a24693f19fb2e2a50ecd1bf.

This matters because the manuscript already contains several earlier continuations. Its Gaussian parity reductions, depth-four reciprocal identities, modular row convergence bounds, and global Stieltjes zero theorem are established material. They are used as context or reproved where needed, rather than presented as new discoveries.

The present work follows three distinct threads. The first concerns the exact integer structure of a specified relation matrix. The second turns empirical CM products into consequences of the arithmetic of modular functions and a classical period formula. The third determines positive real zeros by combining global analytic bounds with finite certificates. These threads share a methodological point: a proposed identity or zero count becomes substantially more useful when its precise proof mechanism and its limits are stated.

1.1 What is proved, and what remains open

The following map separates the main claims and their evidence.

Claim	Result and proof mechanism	Location
Odd-weight formal ranks	The rank conjecture holds for every odd weight; integer polynomial substitutions also determine saturation of the row lattice.	Theorem 2.1
Even-weight formal system	Full column rank; all nonunit invariant factors equal 2.	Theorem 2.1
Mixed-color values	A finite product certificate evaluates $\Im \text{Li}_{w-1,1}(i, -1)$ at every even weight.	Theorem 3.1
CM class products	An all-even-weight resultant formula; five baseline weight-four products and three displayed weight-six extensions.	Theorem 4.2
CM genus ratios	The recorded ratios and weight-six extensions are proved; one number field of degree at most 12 contains every even-weight ratio for a fixed character.	Theorems 5.2, 5.4
Fifth-index real zeros	Exactly three at derivative orders one and two, and five thereafter, all simple.	Theorem 7.3
Lerch deformation	Complete small-parameter classification, alternating branch directions, and a proved double-zero bifurcation.	Theorems 8.2, 8.4, 8.6

The Gaussian matrix theorem is about *formal relations*. It does not compute the dimension of a space of actual transcendental numbers. The CM identities use classical period theory; the contribution is a fully specified formula in the manuscript's normalization and its exact applications. The fifth-index zero count uses a computer-assisted finite lemma with explicit rational arithmetic, not an unproved numerical root scan. The small-parameter and bifurcation results have analytic proofs; decimal bifurcation coordinates are supplementary diagnostics.

We do not settle the manuscript's S_4 mixed-relation conjecture, general rational-argument weight-four ladders, or its highest-weight algebraic ladder claims. In particular, an obstruction to one finite relation system is not an obstruction to all functional equations. Priority claims here

are relative to the pinned manuscript. The literature survey identifies the classical ingredients and relevant existing methods, but does not justify asserting that every displayed consequence is absent from the entire historical literature.

1.2 Polylogarithms, convergence, and color conventions

We use the strict nested-sum convention

$$\mathrm{Li}_{a,b}(x,y) = \sum_{m>n\geq 1} \frac{x^m y^n}{m^a n^b}, \quad \mathrm{Li}_s(z) = \sum_{m\geq 1} \frac{z^m}{m^s}. \quad (1.1)$$

This fixes the distinction between colors (x,y) and the adjacent-letter coordinates in an iterated integral. At roots of unity the values are ordinary convergent boundary values where the series converges; the one divergent imaginary coordinate that occurs in the formal system is explicitly omitted. Its regularized cancellation rows are exactly those specified by the baseline. No additional regularization convention is silently introduced.

For positive integer s , write

$$\beta(s) = \sum_{m\geq 0} \frac{(-1)^m}{(2m+1)^s} = \Im \mathrm{Li}_s(i), \quad \eta_{\mathrm{alt}}(s) = \sum_{m\geq 1} \frac{(-1)^{m-1}}{m^s}.$$

Thus $\eta_{\mathrm{alt}}(1) = \log 2$ and $\eta_{\mathrm{alt}}(s) = (1 - 2^{1-s})\zeta(s)$ for $s > 1$. The subscript distinguishes this alternating Dirichlet series from the Dedekind eta function $\eta(\tau)$ in the CM sections.

The colored shuffle and stuffle framework is standard. Zhao's work [2] distinguishes the various classes of standard relations, while Yuan–Zhao [3] provide closely relevant formal double-space and polynomial constructions. We use this context to locate the exact matrix theorem, without identifying its particular quotient with a larger motivic or numerical period space.

1.3 Analytic conventions

The generalized Stieltjes constants are defined by

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n\geq 0} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n, \quad a > 0. \quad (1.2)$$

A superscript (k) on γ_n means an a derivative. A derivative ∂_s of a zeta or Lerch function always means a derivative in its complex order. We keep these two operations separate. The elementary Lerch endpoint is $\rho = 0$; the classical Hurwitz endpoint is $\rho = 1$. The coefficients themselves need not have a convergent series at the classical endpoint, but their parameter derivatives of positive order do. Definitions and the standard Mellin representations are recorded in the DLMF [5, 4].

All root counts are on the open positive real axis. A count with *multiplicity* is explicitly identified as such. When a theorem asserts simplicity, the multiplicity count and the number of distinct zeros agree. This distinction is essential at the elementary endpoint and at a bifurcation.

2 Exact ranks and Smith forms for the level-four product system

The manuscript's odd-weight rank conjecture concerns a fully specified matrix of depth-two relations. It does not concern the dimension of the space of numerical periods. We prove that conjecture, determine the corresponding matrices in even weight, and compute their integral Smith forms. The reason for the simple answer is a polynomial involution which becomes a permutation of monomials after an integral change of variables.

2.1 The matrix and the result

For $a + b = w$, write formal coordinates

$$x_{a,b}^{r,s}, \quad r, s \in \mathbb{Z}/4\mathbb{Z},$$

representing $\Im \text{Li}_{a,b}(i^r, i^s)$. Set them equal to zero when r, s are both even, and impose $x_{a,b}^{r,s} = -x_{a,b}^{-r,-s}$. The six canonical color pairs are

$$(0, 1), (1, 0), (1, 1), (1, 2), (1, 3), (2, 1). \quad (2.1)$$

Remove $x_{1,w-1}^{0,1}$, the single divergent imaginary coordinate after conjugation. Thus the coordinate count is $6w - 7$.

For $p + q = w$ and $x = i^r, y = i^s$, the two product rows have left sides

$$S_{p,q}^{r,s} = x_{p,q}^{r,s} + x_{q,p}^{s,r}, \quad (2.2)$$

$$H_{p,q}^{r,s} = \sum_{j=0}^{p-1} \binom{q+j-1}{j} x_{q+j,p-j}^{s,r-s} + \sum_{j=0}^{q-1} \binom{p+j-1}{j} x_{p+j,q-j}^{r,s-r}. \quad (2.3)$$

Their analytic right sides are respectively

$$\Im(\text{Li}_p(x) \text{Li}_q(y) - \text{Li}_w(xy)), \quad \Im(\text{Li}_p(x) \text{Li}_q(y)).$$

When both single factors are finite, retain both rows. When exactly one factor is $\text{Li}_1(1)$, retain the difference $S - H$, in which the divergent coordinate cancels. Its right side is $-\Im \text{Li}_w(xy)$. Discard zero rows and retain the remaining integer coefficient matrix A_w . Repeated rows do not affect the assertions below. This is precisely the matrix in the manuscript's Conjecture `signed:conj:rank`. Let A_w^{-g} be the matrix obtained by deleting the $w - 1$ columns $x_{a,w-a}^{1,0}$; it has $5w - 6$ columns.

Theorem 2.1 (All-weight ranks and integral invariant factors). *For every odd $w \geq 3$,*

$$\text{rank } A_w = 5w - 6, \quad \text{rank } A_w^{-g} = \frac{9w - 11}{2}. \quad (2.4)$$

Every nonzero Smith invariant of either integer matrix is one. For every even $w \geq 4$, both matrices have full column rank, and their nonzero Smith invariants are

$$\text{SNF}(A_w) = \text{diag}(1^{[5w-7]}, 2^{[w]}), \quad (2.5)$$

$$\text{SNF}(A_w^{-g}) = \text{diag}(1^{[(9w-12)/2]}, 2^{[w/2]}). \quad (2.6)$$

Here the bracketed exponent denotes multiplicity; extra zero rows in a rectangular Smith form are suppressed.

2.2 A unit pivot restores the omitted coordinate

Restore $t = x_{1,w-1}^{0,1}$ and impose both homogeneous rows (2.2)–(2.3) for every color and index pair. Denote the resulting auxiliary matrix by M_w . The excluded stuffle row for $(p, q, r, s) = (1, w - 1, 0, 1)$ is

$$t + x_{w-1,1}^{1,0} = 0. \quad (2.7)$$

Every excluded stuffle row with a nonzero imaginary part is this row up to sign and interchange of factors. The other excluded stuffle rows vanish under the imposed color conventions. Adding

(2.7) to the baseline difference rows therefore recovers all the missing individual stuffle and shuffle rows by integral row combinations. Conversely, their differences recover the baseline rows. A finite-product row cannot contain t : an outer index one and outer color zero in a shuffle term require the corresponding input factor to be $\text{Li}_1(1)$.

The coefficient of t in (2.7) is one, so eliminating it is unimodular. The row-module quotients for M_w and A_w are isomorphic; the Smith form of M_w has exactly one additional unit. Their homogeneous nullspaces over any field are isomorphic, with $t = -x_{w-1,1}^{1,0}$. After deleting the Gaussian columns, the same argument applies with those coordinates set to zero and $t = 0$.

This auxiliary construction is a statement about integer coefficient matrices. It assigns no analytic finite value to the divergent coordinate and introduces no new analytic relation.

2.3 Six polynomials reduce to two involutions

Put $n = w - 2$ and encode a vector of coordinates by homogeneous polynomials

$$f_{r,s}(X, Y) = \sum_{a=1}^{w-1} x_{a,w-a}^{r,s} X^{a-1} Y^{w-a-1}.$$

The complete homogeneous product equations are exactly

$$f_{r,s}(X, Y) + f_{s,r}(Y, X) = 0, \quad (2.8)$$

$$f_{s,r-s}(X + Y, X) + f_{r,s-r}(X + Y, Y) = 0. \quad (2.9)$$

For example, the coefficient of $X^{p-1}Y^{q-1}$ in the second term of (2.9) is

$$\sum_{j=0}^{q-1} \binom{p+j-1}{j} x_{p+j,q-j}^{r,s-r},$$

which is the second binomial sum in (2.3).

Write

$$A = f_{0,1}, \quad B = f_{1,0}, \quad C = f_{1,1}, \quad D = f_{1,2}, \quad E = f_{1,3}, \quad F = f_{2,1}.$$

The stuffle equations become

$$A(X, Y) = -B(Y, X), \quad F(X, Y) = -D(Y, X), \quad (2.10)$$

$$C(X, Y) = -C(Y, X), \quad E(X, Y) = E(Y, X). \quad (2.11)$$

Up to conjugation and interchange of the factors, the nonzero shuffle equations are exhausted by colors $(0, 1), (1, 1), (2, 1), (1, 3)$. With $V = X + Y$, their left sides are respectively

$$E(V, X) + A(V, Y), \quad B(V, X) + B(V, Y), \quad C(V, X) - F(V, Y), \quad -D(V, X) + D(V, Y).$$

The color pairs $(0, 0), (0, 2), (2, 0), (2, 2)$ vanish. Conjugation supplies the pairs $(0, 3), (3, 0), (3, 3), (3, 2), (3, 1), (2, 3)$; the remaining pairs $(1, 0)$ and $(1, 2)$ are factor interchanges. This checks all sixteen color pairs.

Combining these equations with (2.10) gives

$$E(X, Y) = B(X - Y, X), \quad B(X, X - Y) = -B(X, Y), \quad (2.12)$$

$$C(X, Y) = -D(X - Y, X), \quad D(X, X - Y) = D(X, Y). \quad (2.13)$$

Thus A, F, E, C are eliminated and no relation couples B to D .

Let $T(X, Y) = (X, X - Y)$. The remaining conditions (2.11), transported through the substitution $U(X, Y) = (X - Y, X)$, yield the coefficient rows

$$(I + T)B, \quad (I - (-1)^n T)B, \quad (I - T)D, \quad (I + (-1)^n T)D. \quad (2.14)$$

Here T acts by substitution. To check the signs, if $S(X, Y) = (Y, X)$, then $USU^{-1} = -T$, and a degree- n homogeneous polynomial takes the factor $(-1)^n$ under simultaneous sign change.

These are integral presentation operations. Each of the four eliminated polynomials has coefficient one. The substitutions and their inverses have integer coefficients and determinant ± 1 , so they induce unimodular maps of the coefficient lattices. Consequently the four eliminations contribute $4(w - 1)$ Smith units, and the remaining Smith problem is exactly (2.14).

2.4 The integral monomial permutation proves the theorem

Make the unimodular change of variables

$$u = Y, \quad v = X - Y, \quad X = u + v, \quad Y = u. \quad (2.15)$$

The involution T now exchanges u, v . It therefore permutes the integral monomial basis $u^j v^{n-j}$ by $j \leftrightarrow n - j$.

If $n = 2m - 1$ is odd, this permutation has m two-element orbits and no fixed monomial. On each orbit the two B relations in (2.14) coincide and give the primitive row $(1, 1)$. The two D relations coincide and give $(1, -1)$. Each orbit contributes one Smith unit and one free coordinate in each polynomial. Hence the full nullity is $2m = w - 1$, and the nullity with $B = 0$ is $m = (w - 1)/2$. Counting the $4(w - 1)$ eliminated units and subtracting the auxiliary unit (2.7) gives respectively

$$4(w - 1) - 1 + (w - 1) = 5w - 6, \quad 4(w - 1) - 1 + \frac{w - 1}{2} = \frac{9w - 11}{2}$$

nonzero Smith units. This proves (2.4) and its integral strengthening.

If $n = 2h$ is even, there are h two-element orbits and one fixed monomial. Both plus and minus relations occur for each of B, D . A two-element orbit gives

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

with Smith form $\text{diag}(1, 2)$; a fixed monomial gives the single nonzero row (2) . Each polynomial therefore contributes h units and $h + 1$ twos, with no free coordinate. Adding the eliminated units and removing the auxiliary unit gives

$$4(w - 1) - 1 + (w - 2) = 5w - 7 \quad \text{units}, \quad w \quad \text{twos}$$

for A_w , and

$$4(w - 1) - 1 + \frac{w - 2}{2} = \frac{9w - 12}{2} \quad \text{units}, \quad \frac{w}{2} \quad \text{twos}$$

for A_w^{-g} . This proves (2.5)–(2.6) and completes the proof of Theorem 2.1.

2.5 Primitive witnesses and exact certificate consequences

For odd $n = 2m - 1$ and $0 \leq j < m$, define

$$\begin{aligned} P_j(X, Y) &= Y^j(X - Y)^{n-j} - Y^{n-j}(X - Y)^j, \\ Q_j(X, Y) &= Y^j(X - Y)^{n-j} + Y^{n-j}(X - Y)^j. \end{aligned}$$

The complete integer nullspace is obtained, uniquely, by setting

$$B = \sum_{j=0}^{m-1} b_j P_j, \quad D = \sum_{j=0}^{m-1} d_j Q_j, \quad b_j, d_j \in \mathbb{Z},$$

using (2.10), (2.12), (2.13) for the other colors, and omitting the restored coordinate. Primitivity follows directly from the monomial pairs in the unimodular coordinates (2.15).

Corollary 2.2 (All Gaussian-only consequences). *In odd weight w , the rational row space of A_w supported only on the Gaussian columns has dimension $(w - 1)/2$. It is exactly the span of the $(w - 1)/2$ same-point shuffle rows. Every integer vector in this space is a unique integer combination of those rows.*

Proof. The kernel of projection of the row space to non-Gaussian columns has dimension $\text{rank } A_w - \text{rank } A_w^{-g} = (w - 1)/2$. The same-point shuffle rows with $p + q = w$, $1 \leq p \leq m = (w - 1)/2$, have Gaussian coefficients

$$R_{p,a} = \binom{a-1}{p-1} + \binom{a-1}{w-p-1}.$$

Their first m columns form the triangular Pascal matrix $\binom{a-1}{p-1}$, with determinant one. They are independent and span the indicated rational space. The same unit minor shows that an integer vector in this space has integer expansion coefficients. $\square$

The Smith forms have two further direct consequences. In odd weight, every integer coefficient vector in the rational row span has an integer row certificate. In even weight, twice every coordinate vector has an integer row certificate. Thus every convergent imaginary level-four double in even weight reduces with coefficients in $\frac{1}{2}\mathbb{Z}$ to the raw finite-product and single-value right sides of the stated system. This denominator assertion is relative to that raw vocabulary; rewriting the single values in powers of π , zeta values, and beta values can introduce their usual additional rational factors.

At weight five a particularly short obstruction is obtained by choosing $B = 0, D = X^3$. Then

$$C = -(X - Y)^3, \quad F = -Y^3, \quad A = E = 0.$$

The six nonzero coordinates of this integer kernel vector are

$$x_{4,1}^{1,2} = 1, \quad x_{1,4}^{2,1} = -1, \quad x_{4,1}^{1,1} = -1, \quad x_{3,2}^{1,1} = 3, \quad x_{2,3}^{1,1} = -3, \quad x_{1,4}^{1,1} = 1.$$

The manuscript's target $3g_{4,1} + 3g_{3,2} + 9g_{2,3} + 7u_{4,1}$ pairs with this vector to give 7. This proves again that the proposed mixed relation is absent from the specified linear product system. It does not disprove the numerical identity or preclude its derivation from additional relations.

Scope and verification. The theorem proves the manuscript's conjecture and a stronger integral normal form. It makes no assertion that this relation system contains all standard polylogarithm relations, and no assertion of arithmetic independence. Formal double spaces, polynomial relation methods, and colored regularization are established tools; appropriate context is provided by Yuan–Zhao [3] and Zhao [2]. The new contribution here is the exact normal form of the particular matrix family under study. The independent companion script reconstructs the displayed rows, recovers the baseline 92×23 matrix size at weight five, checks ranks and primitive nullspace residuals through weight thirteen, and checks both Smith forms through weight twelve. The general result rests on the integral proof above, rather than on extrapolation from these finite checks.

3 An explicit mixed-color identity at every even weight

The polynomial reduction also gives a concrete infinite family of identities. The following proof stays entirely within three convergent color families, so it needs no auxiliary divergent coordinate. Write

$$u_{a,b} = \Im \operatorname{Li}_{a,b}(i, -1), \quad \beta(s) = \Im \operatorname{Li}_s(i), \quad \eta_{\text{alt}}(s) = -\operatorname{Li}_s(-1).$$

Thus $\eta_{\text{alt}}(1) = \log 2$, and $\eta_{\text{alt}}(s) = (1 - 2^{1-s})\zeta(s)$ for integers $s \geq 2$.

Theorem 3.1 (One trailing index in the mixed Gaussian family). *For every even integer $w \geq 2$,*

$$u_{w-1,1} = \frac{w}{2} \beta(w) - \log 2 \beta(w-1) - \sum_{\substack{1 \leq a < w \\ a \text{ modd}}} \eta_{\text{alt}}(a) \beta(w-a) + 2^{1-w} \beta(1) \eta_{\text{alt}}(w-1). \quad (3.1)$$

In particular,

$$u_{1,1} = \beta(2) - \frac{3\pi}{8} \log 2, \quad (3.2)$$

$$u_{3,1} = 2\beta(4) - \frac{\pi^3}{16} \log 2 - \frac{21\pi}{128} \zeta(3), \quad (3.3)$$

$$u_{5,1} = 3\beta(6) - \frac{5\pi^5}{768} \log 2 - \frac{3\pi^3}{128} \zeta(3) - \frac{465\pi}{2048} \zeta(5). \quad (3.4)$$

Proof. Put $n = w - 2$ and define degree- n homogeneous polynomials

$$\begin{aligned} C(X, Y) &= \sum_{a+b=w} \Im \operatorname{Li}_{a,b}(i, i) X^{a-1} Y^{b-1}, \\ D(X, Y) &= \sum_{a+b=w} u_{a,b} X^{a-1} Y^{b-1}, \\ F(X, Y) &= \sum_{a+b=w} \Im \operatorname{Li}_{a,b}(-1, i) X^{a-1} Y^{b-1}, \\ P(X, Y) &= \sum_{a+b=w} \Im(\operatorname{Li}_a(-1) \operatorname{Li}_b(i)) X^{a-1} Y^{b-1}, \\ Q(X, Y) &= \sum_{a+b=w} \Im(\operatorname{Li}_a(i) \operatorname{Li}_b(i)) X^{a-1} Y^{b-1}, \\ R(X, Y) &= \sum_{a+b=w} \Im(\operatorname{Li}_a(i) \operatorname{Li}_b(-i)) X^{a-1} Y^{b-1}, \\ H(X, Y) &= \sum_{j=0}^n X^j Y^{n-j}. \end{aligned}$$

All indices a, b in these sums are positive. Every single factor is finite, including those of index one. The colored stuffle and shuffle identities give exactly

$$F(X, Y) + D(Y, X) = P(X, Y) + \beta(w)H(X, Y), \quad (3.5)$$

$$C(X, Y) + C(Y, X) = Q(X, Y), \quad (3.6)$$

$$C(X + Y, X) - F(X + Y, Y) = P(X, Y), \quad (3.7)$$

$$D(X + Y, Y) - D(X + Y, X) = R(X, Y). \quad (3.8)$$

The signs in the first and third equations use respectively $\Im \text{Li}_w(-i) = -\beta(w)$ and $\Im \text{Li}_{a,b}(-1, -i) = -\Im \text{Li}_{a,b}(-1, i)$.

Let $d = D(1, 0) = u_{w-1,1}$. Equation (3.8) at $(X, Y) = (0, 1)$ says $D(1, 1) = d + R(0, 1)$. Using (3.7) at $(0, 1)$ and (3.5) at $(1, 1)$ gives

$$C(1, 0) = -D(1, 1) + P(1, 1) + P(0, 1) + (w - 1)\beta(w).$$

Since n is even, $F(0, -1) = F(0, 1)$. The same equations at $(1, -1)$ and $(0, 1)$ give

$$C(0, 1) = -d + P(0, 1) + P(1, -1) + \beta(w).$$

Adding and using (3.6) yields the exact identity

$$2d = P(1, 1) + P(1, -1) + 2P(0, 1) + w\beta(w) - Q(1, 0) - R(0, 1). \quad (3.9)$$

Because $a + b = w$ is even, a and b have the same parity. Therefore

$$P(1, 1) + P(1, -1) = -2 \sum_{\substack{1 \leq a < w \\ a \text{ modd}}} \eta_{\text{alt}}(a)\beta(w - a), \quad P(0, 1) = -\log 2 \beta(w - 1).$$

Finally,

$$\begin{aligned} Q(1, 0) + R(0, 1) &= \Im [\text{Li}_1(i)(\text{Li}_{w-1}(i) + \text{Li}_{w-1}(-i))] \\ &= 2\beta(1)\Re \text{Li}_{w-1}(i) = -2^{2-w}\beta(1)\eta_{\text{alt}}(w - 1). \end{aligned}$$

Substitution in (3.9) and division by two prove (3.1). The displayed examples use $\beta(1) = \pi/4$, $\beta(3) = \pi^3/32$, and $\beta(5) = 5\pi^5/1536$. $\square$

For an independent numerical check, the defining iterated integral gives

$$u_{w-1,1} = \Im \left[\frac{1}{(w-2)!} \int_0^1 \frac{-i \log(1+it)}{1-it} (-\log t)^{w-2} dt \right]. \quad (3.10)$$

The path meets no singularity. Expansion in the open disk, followed by the ordinary boundary limit, verifies this representation. Its integrand is $O(t|\log t|^{w-2})$ at zero, so it is integrable even at $w = 2$. Independent quadrature of (3.10) agrees with (3.1) at weights 2, 4, 6, 8, 10 to the working precision of seventy decimal digits. In particular

$$u_{3,1} \approx 0.01508330704224296705, \quad u_{5,1} \approx 0.001899710391767247024.$$

The proof is the finite polynomial calculation above; the quadrature is a diagnostic. Classical colored parity theorems predict reducibility in this sector. The formula supplies a compact explicit family and a direct proof; no claim of worldwide priority is required for that consequence.

A short integer row certificate. Using the left-side row notation of (2.2)–(2.3), the entire family has the explicit coefficient certificate

$$2x_{w-1,1}^{1,2} = \sum_{p=1}^{w-1} S_{p,w-p}^{2,1} + H_{1,w-1}^{2,1} + \sum_{p=1}^{w-1} (-1)^{w-p-1} H_{p,w-p}^{2,1} + S_{1,w-1}^{2,1} - S_{w-1,1}^{1,1} - H_{1,w-1}^{1,3}. \quad (3.11)$$

All rows on the right come from finite products. This identity is understood after the manuscript's color-conjugation identifications. Its coefficients are integers in $\{-1, 1\}$ before equal rows are collected, and it uses $2w + 2$ listed row occurrences. The companion script checks the exact coefficient cancellation through weight twenty.

4 Exact CM row products from class polynomials

The manuscript's complex-multiplication chapter records striking products of weight-four polygamma row sums at discriminants $-15, -20, -23, -39, -47$. Its displayed numerical residuals establish useful evidence, but do not explain the prime factors or provide proofs of the gamma products. The following argument supplies that missing step. It also evaluates the corresponding products at every even weight by a finite computation with a Hilbert class polynomial.

The analytic ingredients are classical: the Kronecker limit formula, the Lerch–Chowla–Selberg formula, and the algebra of modular forms. The contribution here is an explicit, uniformly normalized consequence for the manuscript's row sums, accompanied by certified class-polynomial data and new displayed weight-six evaluations. We do not claim that the underlying CM period theory is new.

4.1 Conventions and the main theorem

Let $D = -d < -4$ be a fundamental discriminant, let $\chi_D(r) = (D/r)$ be its primitive quadratic Dirichlet character, and let $h = h(D)$. Take the standard reduced primitive positive definite forms

$$\mathcal{Q}_D = \{(a, b, c) : b^2 - 4ac = D, |b| \leq a \leq c, b \geq 0 \text{ if } |b| = a \text{ or } a = c\}.$$

There is one representative for each proper ideal class. Write

$$\begin{aligned} \tau_Q &= \frac{-b + i\sqrt{d}}{2a}, & A_D &= \prod_{Q \in \mathcal{Q}_D} a, & P_D &= \prod_{\substack{1 \leq r < d \\ \chi_D(r)=1}} \Gamma(r/d), \\ H_D(X) &= \prod_{Q \in \mathcal{Q}_D} (X - j(\tau_Q)), & C_d &= \Phi_d(1), & \varphi_d &= \varphi(d). \end{aligned} \quad (4.1)$$

Thus $H_D \in \mathbb{Z}[X]$ is monic; $C_d = p$ if d is a positive power of a prime p , and $C_d = 1$ otherwise. All fractional powers of positive real numbers below denote their positive values.

For even $w \geq 4$, use exactly the lattice normalization of the manuscript:

$$G_w(\tau) = \sum_{(m,n) \neq (0,0)} (m + n\tau)^{-w} = 2\zeta(w)E_w(\tau). \quad (4.2)$$

The modular discriminant is $\Delta(\tau) = \eta(\tau)^{24}$; in particular $j = E_4^3/\Delta$ and $j - 1728 = E_6^2/\Delta$.

Lemma 4.1 (The polynomial attached to an even weight). *For every even integer $w \geq 4$, there is a unique monic polynomial $F_w \in \mathbb{Q}[X]$ of degree w such that*

$$\frac{E_w(\tau)^{12}}{\Delta(\tau)^w} = F_w(j(\tau)). \quad (4.3)$$

It is effectively computable using exact rational Fourier coefficients.

Proof. The left side is a weight-zero modular function, holomorphic on the upper half-plane since Δ has no zeros there, with a pole of order w at the cusp and no other poles. The field of modular functions is $\mathbb{C}(j)$, and a modular function with no finite poles is a polynomial in j . Since $E_w = 1 + O(q)$, $\Delta = q + O(q^2)$, and $j = q^{-1} + 744 + O(q)$, this polynomial is monic of degree w . Successively removing its Laurent coefficients at $q^{-w}, \dots, q^0$ computes all coefficients. The input coefficients are rational, hence so are the polynomial coefficients. Equivalently, one can work in the exact graded algebra $\mathbb{Q}[E_4, E_6]$. $\square$

Set

$$\mathcal{R}_{D,w} = \text{Res}_X(H_D(X), F_w(X)).$$

Our resultant convention is $\text{Res}(H_D, F_w) = \prod_Q F_w(j(\tau_Q))$, since H_D is monic. No unidentified sign or leading coefficient is implicit.

Theorem 4.2 (CM product formula at every even weight). *With the preceding definitions, for every even $w \geq 4$,*

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)| = \left(\frac{|B_w|}{w!} \right)^h \left(\frac{A_D}{d^h} \right)^{w/2} C_d^{w/4} (2\pi)^{w(2h-\varphi_d)/4} P_D^w |\mathcal{R}_{D,w}|^{1/12}. \quad (4.4)$$

Here the Bernoulli numbers satisfy $B_2 = 1/6$. The formula includes a zero product: it vanishes precisely when $\mathcal{R}_{D,w} = 0$.

The absolute values are essential in a uniform statement. Selecting one standard representative for each class need not make the product of positive-weight modular forms real. Section 4.5 gives the phase information needed for the manuscript's examples.

4.2 Derivation of the period factor

Put

$$R_D = \prod_{r=1}^{d-1} \Gamma(r/d)^{\chi_D(r)}.$$

We first establish the precise normalization

$$\prod_{Q \in \mathcal{Q}_D} |\Delta(\tau_Q)| = \frac{A_D^6 R_D^6}{(2\pi d)^{6h}}. \quad (4.5)$$

This is the fundamental-discriminant case of the classical Lerch–Chowla–Selberg formula. For a modern primary treatment with all normalization factors, see Barquero-Sanchez et al. [9] and Cohen[10]. Here is a derivation that makes the powers of 2, π , and d checkable.

For $\text{Re } s > 1$, define the full Epstein sum

$$Z(\tau, s) = \sum_{(m,n) \neq (0,0)} \frac{(\text{Im } \tau)^s}{|m + n\tau|^{2s}}.$$

The class decomposition of the Dedekind zeta function gives

$$\sum_{Q \in \mathcal{Q}_D} Z(\tau_Q, s) = 2 \left(\frac{\sqrt{d}}{2} \right)^s \zeta(s) L(s, \chi_D). \quad (4.6)$$

Indeed $Q(m, -n) = a|m + n\tau_Q|^2$ (the sign change in n does not affect the full lattice sum), while $a \operatorname{Im} \tau_Q = \sqrt{d}/2$; the factor 2 is the number of units, since $D < -4$.

The Kronecker limit formula at zero reads

$$Z(\tau, 0) = -1, \quad Z'(\tau, 0) = -2 \log(2\pi \sqrt{\operatorname{Im} \tau} |\eta(\tau)|^2).$$

Also $L(0, \chi_D) = h$, and the finite Hurwitz-zeta expression for L and Lerch's identity give

$$L'(0, \chi_D) = -h \log d + \log R_D.$$

Differentiate (4.6) at zero, using $\zeta(0) = -1/2$ and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$. After collecting terms, one obtains

$$\prod_Q \sqrt{\operatorname{Im} \tau_Q} |\eta(\tau_Q)|^2 = (4\pi \sqrt{d})^{-h/2} R_D^{1/2}. \quad (4.7)$$

Finally, $\prod_Q \operatorname{Im} \tau_Q = (\sqrt{d}/2)^h / A_D$. Dividing (4.7) by the square root of this product and raising to the twelfth power proves (4.5).

Because χ_D is odd, the positive and negative character residues are interchanged by $r \mapsto d - r$. Euler reflection therefore gives

$$R_D = \frac{P_D^2}{\pi^{\varphi_d/2}} \prod_{\chi_D(r)=1} \sin(\pi r/d) = \frac{P_D^2 \sqrt{C_d}}{(2\pi)^{\varphi_d/2}}. \quad (4.8)$$

To verify the last equality, square the sine product: it becomes the product over all reduced residues, because paired sines are equal. The identity $|1 - e^{2\pi i r/d}| = 2 \sin(\pi r/d)$ and the cyclotomic polynomial evaluated at 1 give its square as $C_d/2^{\varphi_d}$. Every sine is positive, fixing the square-root sign.

Proof of Theorem 4.2. Multiply (4.3) over the class representatives and take absolute values:

$$\prod_Q |E_w(\tau_Q)| = \left(\prod_Q |\Delta(\tau_Q)| \right)^{w/12} |\mathcal{R}_{D,w}|^{1/12}.$$

Substitute (4.5), then $2\zeta(w) = |B_w|(2\pi)^w/w!$ and (4.8). Directly collecting exponents gives (4.4). No logarithm is taken of E_w , so the same argument applies when a factor vanishes. $\square$

4.3 A finite arithmetic recipe and useful low weights

The general formula is particularly simple in the first few weights:

w	$F_w(X)$	$ \mathcal{R}_{D,w} ^{1/12}$
4	X^4	$ H_D(0) ^{1/3}$
6	$(X - 1728)^6$	$ H_D(1728) ^{1/2}$
8	X^8	$ H_D(0) ^{2/3}$
10	$X^4(X - 1728)^6$	$ H_D(0) ^{1/3} H_D(1728) ^{1/2}$
12	$(X - 432000/691)^{12}$	$ H_D(432000/691) $

For the last row, use $E_{12} = (441E_4^3 + 250E_6^2)/691 = \Delta(j - 432000/691)$. This table is an exact algebraic calculation. In particular, the prime factors of a class product are no longer guessed from a ratio: at weights four and six they are determined by the factorizations of $H_D(0)$ and $H_D(1728)$, together with the explicitly displayed elementary factors.

Specializing the theorem to weight four gives

$$\prod_Q |G_4(\tau_Q)| = \frac{|H_D(0)|^{1/3} A_D^2 C_d}{180^h d^{2h} 2^{\varphi_d} \pi^{\varphi_d - 2h}} P_D^4. \quad (4.9)$$

At weight six it gives

$$\prod_Q |G_6(\tau_Q)| = \frac{A_D^3 C_d^{3/2} |H_D(1728)|^{1/2}}{30240^h d^{3h}} (2\pi)^{3h - 3\varphi_d/2} P_D^6. \quad (4.10)$$

4.4 An exact generator for the modular weight polynomial

Normalize the Eisenstein series to have constant coefficient one:

$$E_w(\tau) = 1 - \frac{2w}{B_w} \sum_{m \geq 1} \sigma_{w-1}(m) q^m, \quad q = e^{2\pi i \tau},$$

and set

$$\Delta = q \prod_{m \geq 1} (1 - q^m)^{24} = \frac{E_4^3 - E_6^2}{1728}, \quad j = \frac{E_4^3}{\Delta}.$$

For each even $w \geq 4$, let $a \in \{0, 1, 2\}$, $b \in \{0, 1\}$, and $c \geq 0$ be the unique integers satisfying

$$w = 4a + 6b + 12c. \quad (4.11)$$

The excluded weight two is precisely the exceptional small weight in this residue prescription.

The level-one graded-ring structure [13] gives a unique monic polynomial $P_w \in \mathbb{Q}[X]$ of degree c such that

$$E_w = E_4^a E_6^b \Delta^c P_w(j). \quad (4.12)$$

This can be computed using only rational arithmetic. Write $P_w(X) = \sum_{r=0}^c p_r X^r$ and expand the basis

$$E_4^{a+3r} E_6^b \Delta^{c-r}, \quad 0 \leq r \leq c.$$

The basis element indexed by r has first term q^{c-r} with coefficient one. Successive coefficient comparison at $q^0, q^1, \dots, q^c$ therefore determines $p_c, p_{c-1}, \dots, p_0$ by unit triangular elimination. Since E_w has constant term one, $p_c = 1$.

Consequently the polynomial in the CM norm formula is explicitly

$$\boxed{F_w(X) = X^{4a} (X - 1728)^{6b} P_w(X)^{12}.} \quad (4.13)$$

Indeed, raising (4.12) to the twelfth power and using $E_4^3 = j\Delta$, $E_6^2 = (j - 1728)\Delta$, and (4.11) gives $E_w^{12}/\Delta^w = F_w(j)$. The degree of F_w is exactly w and its leading coefficient is one.

Finite certificates. The companion generator computes Δ independently from its Euler product and verifies its normalization against $E_4^3 - E_6^2$. For every even w from four through twenty-four, it verifies exactly the coefficients of $q^0, \dots, q^w$ in

$$E_w^{12} = \sum_{r=0}^w [X^r] F_w(X) E_4^{3r} \Delta^{w-r}. \quad (4.14)$$

Both sides are holomorphic modular forms of weight $12w$ on $\mathrm{SL}_2(\mathbb{Z})$. The level-one Sturm bound is w , so these finitely many rational equalities certify the identity. In characteristic zero this bound follows directly from the valence formula [13, Theorem 2.11]: a nonzero weight- $12w$ holomorphic modular form has order at infinity at most w .

The JSON artifact stores the exact rational coefficients, the compact factorization, the degree and monicity checks, and the finite coefficient certificates. It contains the full range, including

$$F_{14}(X) = X^8(X - 1728)^6, \quad (4.15)$$

$$F_{16}(X) = X^4 \left(X - \frac{3456000}{3617} \right)^{12}, \quad (4.16)$$

$$F_{18}(X) = (X - 1728)^6 \left(X - \frac{9504000}{43867} \right)^{12}, \quad (4.17)$$

$$F_{20}(X) = X^8 \left(X - \frac{209520000}{174611} \right)^{12}, \quad (4.18)$$

$$F_{22}(X) = X^4(X - 1728)^6 \left(X - \frac{35424000}{77683} \right)^{12}, \quad (4.19)$$

$$F_{24}(X) = \left(X^2 - \frac{340364160000}{236364091} X + \frac{30710845440000}{236364091} \right)^{12}. \quad (4.20)$$

For example,

$$E_{14} = E_4^2 E_6, \quad E_{16} = \frac{1617E_4^4 + 2000E_4E_6^2}{3617} = E_4 \Delta \left(j - \frac{3456000}{3617} \right),$$

which directly explain (4.15)–(4.16). The generator implements a standard modular-form construction, supplying explicit reusable coefficients and proof certificates for the CM recipe.

4.5 The phase and the exceptional class-pair average

For a form strictly inside the reduced domain with $b \neq 0$, the form with $-b$ also occurs. Their G_w values are complex conjugates, so their product is positive unless a factor is zero. Boundary forms with $b = 0$ or $b = a < c$ give real G_4 and G_6 , and these values are positive. For G_4 this follows from its positive value high on each boundary ray, continuity, and the fact that E_4 has no zero there except the hexagonal endpoint; for G_6 the same argument uses the unique elliptic zero at i . The excluded discriminants $-3, -4$ are exactly the exceptional endpoints. These are the usual zero-divisor consequences of the valence formula.

The remaining boundary type is $a = c$ and $0 < b < a$. Then $|\tau_Q| = 1$. Modularity and conjugation imply $\overline{E_4(\tau_Q)} = \tau_Q^4 E_4(\tau_Q)$. Following the unit arc from i and using the absence of an interior zero fixes the sign and yields

$$G_4(\tau_Q) = -\tau_Q^{-2} |G_4(\tau_Q)| = \left(\frac{d - b^2}{4a^2} - \frac{ib\sqrt{d}}{2a^2} \right) |G_4(\tau_Q)|. \quad (4.21)$$

Thus an unqualified equality between a raw class product and its positive gamma norm is not valid for arbitrary discriminants. Recent detailed work on the analogous eta-phase issue is Cohen[10]; the elementary modular calculation above suffices for the present cases.

For $D = -15$ the nonprincipal form is $(2, 1, 2)$. Consequently

$$G_4((-1 + i\sqrt{15})/4) = \frac{7 - i\sqrt{15}}{8} |G_4((-1 + i\sqrt{15})/4)|.$$

The manuscript uses its real part, which is $7/8$ of the absolute value. This explains its factor 7 exactly; averaging a conjugate pair must not be confused with taking one class representative in a norm.

4.6 Proofs of the manuscript's five class products

The needed class-polynomial constants and leading-coefficient products are as follows. The supplied interval certificate proves these are the exact Hilbert class polynomials, independently of numerical recognition.

D	h	A_D	$H_D(0)$	$H_D(1728)$
-15	2	2	$-3^6 5^3 11^3$	$3^6 7^4 11^2$
-20	2	2	$-2^{12} 5^3 11^3$	$-2^{16} 11^2 19^2$
-23	3	4	$5^9 11^3 17^3$	$7^6 11^4 19^2 23$
-39	4	12	$3^{15} 17^3 23^3 29^3$	$3^{12} 7^8 19^4 23^2$
-47	5	36	$5^{15} 11^6 23^3 29^3$	$11^6 19^4 23^4 31^2 43^2 47$

The signs and factorizations in this table are exact integer arithmetic; they are also recorded in the machine receipt.

For $D = -23, -39, -47$ there are no unit-arc boundary representatives. The preceding positivity argument removes the absolute values in (4.9). It gives precisely

$$\prod_{Q \in \mathcal{Q}_{-23}} G_4(\tau_Q) = \frac{11 \cdot 17 P_{-23}^4}{2^{24} 3^6 23^5 \pi^{16}}, \quad (4.22)$$

$$\prod_{Q \in \mathcal{Q}_{-39}} G_4(\tau_Q) = \frac{17 \cdot 23 \cdot 29 P_{-39}^4}{2^{28} 3^9 5^4 13^8 \pi^{16}}, \quad (4.23)$$

$$\prod_{Q \in \mathcal{Q}_{-47}} G_4(\tau_Q) = \frac{11^2 \cdot 23 \cdot 29 P_{-47}^4}{2^{52} 3^6 47^9 \pi^{36}}. \quad (4.24)$$

These formulas upgrade the three previously empirical product evaluations to consequences of established analytic identities and exact CM data. No gamma-period independence assertion is used.

For $D = -15$, the extra factor $7/8$ from (4.21) gives, with $\tau_1 = (1 + i\sqrt{15})/2$ and $\hat{G}_4 = \text{Re } G_4((1 + i\sqrt{15})/4)$,

$$G_4(\tau_1) \hat{G}_4 = \frac{77 P_{-15}^4}{2^{13} 3^6 5^5 \pi^4}, \quad P_{-15} = \Gamma(1/15) \Gamma(2/15) \Gamma(4/15) \Gamma(8/15). \quad (4.25)$$

For $D = -20$, (4.9) first gives

$$\prod_Q G_4(\tau_Q) = \frac{11 P_{-20}^4}{2^{14} 3^4 5^5 \pi^4}.$$

The multiplication and reflection formulas yield

$$\frac{\Gamma(1/20)\Gamma(9/20)}{\Gamma(3/20)\Gamma(7/20)} = 4 \cdot 5^{1/4} \sin(3\pi/20) \sin(7\pi/20) = 5^{1/4} \phi,$$

where $\phi = (1 + \sqrt{5})/2$. For the first equality, apply the fivefold gamma multiplication formula at $1/20$ and reflect its $13/20$ and $17/20$ factors. Hence $P_{-20} = 5^{1/4} \phi \Gamma(3/20)^2 \Gamma(7/20)^2$, proving

$$G_4(i\sqrt{5})G_4((-1 + i\sqrt{5})/2) = \frac{11\phi^4\Gamma(3/20)^8\Gamma(7/20)^8}{2^{14}3^45^4\pi^4}. \quad (4.26)$$

4.7 Three explicit weight-six extensions

The same exact class data now produce formulas that were not displayed in the baseline manuscript:

$$\prod_{Q \in \mathcal{Q}_{-23}} G_6(\tau_Q) = \frac{11^2 \cdot 19 P_{-23}^6}{2^{33} 3^9 5^3 23^7 \pi^{24}}, \quad (4.27)$$

$$\prod_{Q \in \mathcal{Q}_{-39}} G_6(\tau_Q) = \frac{19^2 \cdot 23 P_{-39}^6}{2^{38} 3^{15} 5^4 13^{12} \pi^{24}}, \quad (4.28)$$

$$\prod_{Q \in \mathcal{Q}_{-47}} G_6(\tau_Q) = \frac{11^3 19^2 23^2 \cdot 31 \cdot 43 P_{-47}^6}{2^{73} 3^9 5^5 7^5 47^{13} \pi^{54}}. \quad (4.29)$$

They follow by inserting the factorizations of $H_D(1728)$ in (4.10); positivity was already established. For example, at $D = -23$ the square root is $7^3 11^2 19 \sqrt{23}$, and $C_{23}^{3/2} = 23^{3/2}$. Their product contributes 23^2 , which changes 23^9 in the denominator to 23^7 . This calculation is a useful check on the normalization.

Every factor here is a polygamma row sum. For example,

$$G_6(\tau) = 2\zeta(6) + \frac{1}{60} \sum_{n \geq 1} \left[\psi^{(5)}(n\tau) + \psi^{(5)}(1 - n\tau) \right].$$

Consequently (4.27)–(4.29) are exact identities for finite products of exponentially convergent polygamma-row combinations. Weight twelve gives a further family whose entire arithmetic factor is the rational number $|H_D(432000/691)|$.

4.8 How the class-polynomial data are certified

The program `cm_norms.py` enumerates reduced forms by exact integer arithmetic. It then evaluates all singular moduli in rectangular complex intervals using 120 decimal digits of directed interval arithmetic and 70 terms of the Lambert series

$$E_4 = 1 + 240 \sum_{n \geq 1} \frac{n^3 q^n}{1 - q^n}, \quad E_6 = 1 - 504 \sum_{n \geq 1} \frac{n^5 q^n}{1 - q^n}, \quad j = \frac{1728 E_4^3}{E_4^3 - E_6^2}.$$

The interval engine encloses π , $\sqrt{d}$, and the complex exponential. Its output endpoints are finite binary rationals; the program converts them to exact Python fractions before every final containment comparison.

Reduced points satisfy $\text{Im } \tau \geq \sqrt{3}/2$, hence $|q| \leq e^{-\pi\sqrt{3}} < 1/200$. This last strict inequality needs no floating-point assumption: use $\pi > 314/100$, $\sqrt{3} > 173/100$, their product greater than

27/5, and the rational partial sum $\sum_{k=0}^{15} (27/5)^k / k! > 200$. For $r = 1/200$ and $p = 3, 5$, the omitted tail satisfies

$$\left| \sum_{n>N} \frac{n^p q^n}{1 - q^n} \right| \leq \frac{(N+1)^p r^{N+1} \mathcal{A}_p(r)}{(1-r)^{p+2}}, \quad (4.30)$$

where

$$\mathcal{A}_3(r) = 1 + 4r + r^2, \quad \mathcal{A}_5(r) = 1 + 26r + 66r^2 + 26r^3 + r^4.$$

Indeed $n = N + 1 + j \leq (N + 1)(j + 1)$, and $\sum_{j \geq 0} (j + 1)^p r^j = \mathcal{A}_p(r) / (1 - r)^{p+1}$. The code adds the resulting error bound in both real and imaginary coordinates before forming j or multiplying the class polynomial.

Classical CM theory proves in advance that each coefficient of H_D is an integer. The interval product encloses every coefficient in an interval strictly contained in its asserted integer plus $(-1/2, 1/2)$; its imaginary interval contains zero. Thus the unique possible integer is proved, rather than merely recognized to many digits. The maximum real interval widths were

$D = -15$	$D = -20$	$D = -23$	$D = -39$	$D = -47$
$1.36 \cdot 10^{-109}$	$3.95 \cdot 10^{-108}$	$1.83 \cdot 10^{-103}$	$2.80 \cdot 10^{-95}$	$1.51 \cdot 10^{-88}$

The displayed bounds have been rounded upward. Exact interval endpoints, all polynomial coefficients, and the integer factorizations are saved in `cm_norms_receipt.json`.

An additional 25 numerical comparisons test the gamma norm formula at weights 4, 6, 8, 10, 12 for each of the five discriminants. They have relative residuals below $1.1 \cdot 10^{-119}$ at 120 digits. These comparisons are regression checks: they are not interval proofs of the gamma identities, whose proof is Theorem 4.2. Similarly, the certified-polynomial computation is a numerical proof conditional on the standard CM integrality theorem and on the interval library's directed-rounding implementation; it is not a Lean formalization. This evidence boundary is explicit and reproducible.

4.9 Extension to arbitrary quadratic orders

There is also a direct version for $D = D_0 f^2$, where D_0 is fundamental, including the unit exceptions $D_0 = -3, -4$. Let $h = h(D)$ and $d = |D|$, and let $\text{CS}(D_0)$ be Cohen's normalized gamma quotient

$$\text{CS}(D_0) = \left(\prod_{r=1}^{|D_0|-1} \Gamma(r/|D_0|)^{\chi_{D_0}(r)} \right)^{w(D_0)/(2h(D_0))}.$$

Define the explicit conductor factor

$$c(D_0, f) = \prod_{p|f} p^{e(p)}, \quad e(p) = \frac{(1 - p^{-v_p(f)})(1 - \chi_{D_0}(p))}{(1 - 1/p)(p - \chi_{D_0}(p))}.$$

The order form of Lerch–Chowla–Selberg[9, 10] gives the following immediate extension of the proof:

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)| = (2\zeta(w))^h A_D^{w/2} \left(\frac{c(D_0, f) \text{CS}(D_0)}{2\pi d} \right)^{hw/2} |\text{Res}(H_D, F_w)|^{1/12}. \quad (4.31)$$

Thus nonmaximal orders require an explicit conductor correction, not a new unidentified period basket. Setting $f = 1$ and $D < -4$ recovers the main theorem.

4.10 Research directions and scope

The theorem suggests several concrete continuations.

1. Extend the exact polynomial F_w generator supplied here and combine it with certified class-polynomial software such as Sutherland’s CRT methods [11]. This extends the product database to large weights and class numbers without integer-relation searches.
2. Determine signed, rather than absolute, products for all weights by tracking the phases of boundary representatives and the zero signs of E_w along the unit arc. The present phase proof covers the displayed weight-four and weight-six examples; a general norm alone does not settle this problem.
3. Use factorizations of $H_D(0)$ and $H_D(1728)$ to compare the prime support with arithmetic formulas for norms of singular-modulus differences. This gives an explanation for the prime patterns observed in the manuscript, beyond using those patterns to build a PSLQ basis.
4. Extend the certified computations to class-by-class algebraic factors and genus-wise products. A full class norm does not by itself prove class ratios. Section 5 supplies the additional genus-character argument for the recorded ratios at $D = -15, -20, -39$; extending it beyond these examples remains a natural direction.
5. Build a uniform certificate format containing the discriminant, reduced forms, exact H_D , exact F_w , resultant, conductor factor, phase convention, gamma residue set, and analytic tail bound. Each claimed identity then has a finite, reviewable provenance chain.

None of these norm computations proves linear or algebraic independence of individual gamma values or individual polygamma rows. Nor does it imply that arbitrary sums of raw positive-weight CM values are class invariants. The result is a precise product theorem with completely specified representative dependence.

5 Genus characters prove the recorded CM ratios

A class product and a class ratio contain different information. The preceding norm formula controls the former. A nontrivial genus character supplies the missing signed product and proves all three recorded weight-four ratios at discriminants $-15, -20, -39$. The same calculation gives a theorem for every even weight and three explicit weight-six ratios.

The analytic mechanism is classical: twist the Kronecker limit formula by a genus character and use Kronecker’s factorization of its Hecke L -function. A primary reference for the factorization, including its precise character convention, is Murty and Murty, Section 5 [14]; see also Masri[15, eqs. (1.3)–(1.5)]. We give the normalization argument because confusing the ordinary and narrow real-quadratic class numbers, or overlooking the units at discriminants -3 and -4 , changes the answer.

5.1 A genus ratio theorem in arbitrary even weight

Suppose that the fundamental discriminant $D < -4$ has a factorization

$$D = eD_-, \quad e > 1, \quad D_- < 0,$$

where e, D_- are coprime fundamental discriminants. Let $\varepsilon : \text{Cl}(D) \rightarrow \{1, -1\}$ be the corresponding nontrivial genus character. Concretely, if a form Q represents an integer n coprime to D , then

$\varepsilon(Q) = \chi_e(n)$. This value is independent of the represented integer chosen. Retain the reduced representatives τ_Q and the normalizations of Section 4, and put

$$\begin{aligned} A_\varepsilon &= \prod_{Q \in \mathcal{Q}_D} a_Q^{\varepsilon(Q)}, & H_D^\pm(X) &= \prod_{\varepsilon(Q)=\pm 1} (X - j(\tau_Q)), \\ \lambda_- &= L(0, \chi_{D_-}) = \frac{2h(D_-)}{\omega(D_-)}, & \omega(D_-) &= |\mathcal{O}_{D_-}^\times|. \end{aligned} \quad (5.1)$$

Here $h(D_-)$ is the imaginary-quadratic class number. Let $h(e)$ be the *ordinary* real-quadratic class number, and let $u_e > 1$ be the least unit greater than one in the real embedding of $\mathbb{Q}(\sqrt{e})$. Its norm is allowed to be -1 . In particular,

$$\lambda_{-3} = \frac{1}{3}, \quad \lambda_{-4} = \frac{1}{2};$$

replacing either of these values by its imaginary class number would give an incorrect result.

Both genus factors are monic of degree $h(D)/2$, with $H_D = H_D^+ H_D^-$. The two sign sets are unions of genera; in the three examples below they are the two genera themselves. Their coefficients lie in the ring of integers of $\mathbb{Q}(\sqrt{e})$. Indeed, the kernel of ε fixes the quadratic genus extension $\mathbb{Q}(\sqrt{D}, \sqrt{e})$ of $\mathbb{Q}(\sqrt{D})$; the two products are invariant under this kernel. Complex conjugation inverts ideal classes, and inversion preserves ε , so both products are real. Their coefficients therefore belong to the real quadratic subfield. The coefficients are algebraic integers because the singular moduli are algebraic integers.

Theorem 5.1 (Genus ratios at every even weight). *For even $w \geq 4$, suppose that $G_w(\tau_Q) \neq 0$ whenever $\varepsilon(Q) = -1$. Then*

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^{w/2} u_e^{-wh(e)\lambda_-} \left| \frac{\text{Res}(H_D^+, F_w)}{\text{Res}(H_D^-, F_w)} \right|^{1/12}. \quad (5.2)$$

The polynomials F_w and the resultant convention are those of Lemma 4.1. All real roots on the right are positive roots; a vanishing numerator is permitted.

Proof. Write $d = |D|$ and

$$Z(\tau, s) = \sum_{(m,n) \neq (0,0)} \frac{(\text{Im } \tau)^s}{|m + n\tau|^{2s}}.$$

The quadratic-form and ideal-class zeta correspondence, with two units because $D < -4$, gives

$$\sum_Q \varepsilon(Q) Z(\tau_Q, s) = 2 \left(\frac{\sqrt{d}}{2} \right)^s L(s, \chi_e) L(s, \chi_{D_-}). \quad (5.3)$$

Here we used Kronecker's factorization $L_{\mathbb{Q}(\sqrt{D})}(s, \varepsilon) = L(s, \chi_e) L(s, \chi_{D_-})$. For completeness, at an unramified split prime the two quadratic characters have the same sign and give the two equal prime-ideal factors; at an inert prime their signs are opposite and their product is $(1 - p^{-2s})^{-1}$, the factor for the principal ideal (p) . At a ramified prime, coprimality of e, D_- leaves precisely the factor of the unramified Dirichlet character. Thus the Euler products agree in $\text{Re } s > 1$, and continuation gives the identity.

Since χ_e is nontrivial and even, $L(0, \chi_e) = 0$. Differentiate (5.3) at zero and use

$$Z'(\tau, 0) = -2 \log(2\pi\sqrt{y} |\eta(\tau)|^2), \quad y = \text{Im } \tau.$$

The constant disappears because $\sum_Q \varepsilon(Q) = 0$. We obtain

$$\sum_Q \varepsilon(Q) \log(\sqrt{y_Q} |\eta(\tau_Q)|^2) = -\lambda_- L'(0, \chi_e). \quad (5.4)$$

Since $y_Q = \sqrt{d}/(2a_Q)$, its signed logarithmic sum is $\sum_Q \varepsilon(Q) \log y_Q = -\log A_\varepsilon$. Consequently

$$\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 \exp(-12\lambda_- L'(0, \chi_e)). \quad (5.5)$$

To fix the real-quadratic normalization, the completed function

$$\Lambda_e(s) = (e/\pi)^{s/2} \Gamma(s/2) L(s, \chi_e)$$

satisfies $\Lambda_e(s) = \Lambda_e(1-s)$. Its values at zero and one give $2L'(0, \chi_e) = \sqrt{e} L(1, \chi_e)$. The real-quadratic class number formula, in the ordinary-class-number and least-unit convention specified above, is

$$L(1, \chi_e) = \frac{2h(e) \log u_e}{\sqrt{e}}.$$

Thus $L'(0, \chi_e) = h(e) \log u_e$, and (5.5) becomes

$$\boxed{\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}.} \quad (5.6)$$

Finally, apply $|E_w(\tau_Q)|^{12} = |\Delta(\tau_Q)|^w |F_w(j(\tau_Q))|$ to the signed product. The factors $2\zeta(w)$ cancel because the plus and minus character fibers have the same cardinality. The remaining products of $F_w(j)$ are exactly the stated resultants. Taking the positive twelfth root proves the formula. $\square$

In particular, write $R_w(D) = \prod_Q |G_w(\tau_Q)|^{\varepsilon(Q)}$. The two smallest weights give the especially convenient identities

$$R_4(D)^3 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(0)}{H_D^-(0)} \right|, \quad (5.7)$$

$$R_6(D)^2 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(1728)}{H_D^-(1728)} \right|. \quad (5.8)$$

Genus theory identifies the field of the factors $H_D^\pm$; these equations additionally account for the periods of the raw modular forms. Claiming a quadratic raw ratio requires verifying the root extraction. We now perform that step exactly.

5.2 The three recorded ratios, now theorems

The required character and unit data are as follows. A plus or minus sign attached to a form denotes its genus-character value.

D	(e, D_-)	plus forms	A_ε	u_e	λ_-
-15	(5, -3)	(1, 1, 4)	1/2	$(1 + \sqrt{5})/2$	1/3
-20	(5, -4)	(1, 0, 5)	1/2	$(1 + \sqrt{5})/2$	1/2
-39	(13, -3)	(1, 1, 10), (3, 3, 4)	3/4	$(3 + \sqrt{13})/2$	1/3

All the other reduced forms have minus sign. These signs can be checked by represented integers: at $D = -15$, the nonprincipal form represents 2 and $\chi_5(2) = -1$; at $D = -20$, the nonprincipal form represents 7 and $\chi_5(7) = -1$; at $D = -39$, the forms with $a = 2$ have $\chi_{13}(2) = -1$, whereas $(3, 3, 4)$ represents 10 and $\chi_{13}(10) = 1$. The class numbers $h(5) = h(13) = 1$ follow from the real-quadratic Minkowski bound $\sqrt{e}/2 < 2$. The displayed units are fundamental: checking $x^2 - ey^2 = \pm 4$ below their sizes leaves no smaller positive unit.

The genus factors are explicitly

$$H_{-15}^{\pm}(X) = X + \frac{191025 \pm 85995\sqrt{5}}{2}, \quad (5.9)$$

$$H_{-20}^{\pm}(X) = X - 632000 \mp 282880\sqrt{5}, \quad (5.10)$$

$$H_{-39}^{\pm}(X) = X^2 + (165765798 \pm 45975573\sqrt{13})X + \frac{63399280527 \pm 17399806263\sqrt{13}}{2}. \quad (5.11)$$

Multiplication gives the certified H_D polynomials of Section 4. The correspondence between these quadratic-field factors and the displayed genera is also certified: for each CM root, evaluate the other factor in a directed interval enclosure and exclude zero. The attached receipt records the exact rational interval endpoints, so the assignment is not a numerical guess about which radical sign to choose.

Theorem 5.2 (Exact weight-four genus ratios). *With the character data in the table,*

$$R_4(-15) = \frac{(4 + \sqrt{5})^2}{44}, \quad (5.12)$$

$$R_4(-20) = \frac{29 + 12\sqrt{5}}{44}, \quad (5.13)$$

$$R_4(-39) = \frac{103299 + 4440\sqrt{13}}{181424}. \quad (5.14)$$

The first equation is an absolute-value ratio; the precise real-part version in the manuscript is

$$\frac{G_4((1 + i\sqrt{15})/2)}{\operatorname{Re} G_4((1 + i\sqrt{15})/4)} = \frac{2(4 + \sqrt{5})^2}{77}. \quad (5.15)$$

For $D = -20$ and $D = -39$, the positive products represented by $R_4(D)$ are exactly the products and ratios in the manuscript, without an additional phase factor.

Proof. Substitute (5.9)–(5.11) and the table into (5.7). In each of the two quadratic fields, direct multiplication verifies that the cube of the claimed positive number is the right-hand side. For example, for $D = -20$ this is the explicit identity

$$\left(\frac{29 + 12\sqrt{5}}{44}\right)^3 = \frac{1}{64} \left(\frac{1 + \sqrt{5}}{2}\right)^{-6} \left| \frac{-632000 - 282880\sqrt{5}}{-632000 + 282880\sqrt{5}} \right|.$$

The identical finite verification with exponents -4 and -4 , and leading-coefficient ratios $1/2$ and $3/4$, proves the other two cases. The attached exact-arithmetic script verifies all three equalities in $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{13})$ before any decimal evaluation. Positivity makes the cube root unique.

For $D = -15$, the phase calculation of Section 4.5 gives $\operatorname{Re} G_4((1 + i\sqrt{15})/4) = \frac{7}{8}|G_4((1 + i\sqrt{15})/4)|$. Multiplying (5.12) by $8/7$ proves (5.15); translation by one and reflection in the

imaginary axis account for the manuscript's positive-real-part representatives. The phase discussion in the preceding section also shows that the numerator and denominator genus products at -20 and -39 are positive: boundary representatives contribute positive real G_4 , and the two interior representatives at -39 form a conjugate pair. $\square$

This supplies the missing proofs for the ratios originally displayed in the manuscript's Results `cm:res:d15`, `cm:res:d20`, and `cm:res:genus`. Combined with the proved total products, it also determines the individual absolute values at class number two, and the absolute products over each genus at class number four: if T_w is the full positive product, the two genus products are $\sqrt{T_w R_w}$ and $\sqrt{T_w/R_w}$.

5.3 Three explicit weight-six identities

The same genus polynomials have further consequences beyond the recorded weight-four data.

Theorem 5.3 (Weight-six genus ratios). *With the same genera and representatives,*

$$R_6(-15) = \frac{29 + 12\sqrt{5}}{88}, \quad (5.16)$$

$$R_6(-20) = \frac{601 + 252\sqrt{5}}{1672}, \quad (5.17)$$

$$R_6(-39) = \frac{1323 + 324\sqrt{13}}{1472}. \quad (5.18)$$

The last two again hold for the corresponding raw genus products. At discriminant -15 an explicit real-part identity is

$$\frac{G_6((1 + i\sqrt{15})/2)}{\operatorname{Re} G_6((1 + i\sqrt{15})/4)} = \frac{2(29 + 12\sqrt{5})}{121}. \quad (5.19)$$

Proof. Use (5.8). Exact arithmetic in the relevant quadratic field shows that its right-hand sides are, respectively,

$$\begin{aligned} \frac{1561 + 696\sqrt{5}}{7744} &= \left(\frac{29 + 12\sqrt{5}}{88} \right)^2, \\ \frac{678721 + 302904\sqrt{5}}{2795584} &= \left(\frac{601 + 252\sqrt{5}}{1672} \right)^2, \\ \frac{3115017 + 857304\sqrt{13}}{2166784} &= \left(\frac{1323 + 324\sqrt{13}}{1472} \right)^2. \end{aligned}$$

The roots on the right are positive. The valence formula gives the only zero of E_6 in the standard fundamental domain at i ; none of the CM representatives under consideration is this point. Along the boundary rays $\operatorname{Re} \tau = 0$ and $\operatorname{Re} \tau = -1/2$ above the relevant representatives, E_6 is real, has no zero, and tends to one at infinity, so its values are positive. Conjugate interior points again contribute a positive product. This proves the raw-product assertion at $-20, -39$.

On the unit arc $|\tau| = 1$ with $\pi/2 < \arg \tau \leq 2\pi/3$, modularity and conjugation imply that $\tau^3 G_6(\tau)$ is real. It is positive at the hexagonal endpoint, where $\tau^3 = 1$ and $G_6 > 0$, and cannot change sign before i . Thus $G_6(\tau) = \tau^{-3} |G_6(\tau)|$ on this open arc. At $\tau = (-1 + i\sqrt{15})/4$,

$$\tau^{-3} = \frac{11 + 3i\sqrt{15}}{16}, \quad \operatorname{Re} G_6(\tau) = \frac{11}{16} |G_6(\tau)|.$$

Reflection preserves the real part. Multiplying (5.16) by $16/11$ proves (5.19). $\square$

The quadratic norms provide useful exact checks:

$$N(R_4(-15)) = N(R_4(-20)) = \frac{1}{16}, \quad N(R_4(-39)) = \frac{81}{256},$$

and

$$N(R_6(-15)) = N(R_6(-20)) = \frac{1}{64}, \quad N(R_6(-39)) = \frac{729}{4096}.$$

These follow by conjugating $\sqrt{e} \mapsto -\sqrt{e}$ in the displayed exact formulas. The identities are explicit consequences of the classical genus-period mechanism; no assertion of historical priority for individual numerical evaluations is intended.

5.4 A fixed field contains all even-weight genus ratios

The low-weight identities imply a uniform field-degree theorem. For a fixed genus character, increasing the weight introduces no additional algebraic field extension beyond the two seed ratios.

Theorem 5.4 (Uniform algebraic degree of genus ratios). *Under the assumptions of Theorem 5.1, put $K = \mathbb{Q}(\sqrt{e})$ and*

$$\mathcal{K}_{D,\varepsilon} = K(R_4(D), R_6(D)).$$

Then

$$[\mathcal{K}_{D,\varepsilon} : K] \leq 6, \quad [\mathcal{K}_{D,\varepsilon} : \mathbb{Q}] \leq 12,$$

and every defined ratio $R_w(D)$ with even $w \geq 4$ belongs to this same field. The degree bound is independent of w and of the imaginary class number $h(D)$.

Proof. The genus period ratio

$$B_{D,\varepsilon} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}$$

belongs to K . Indeed,

$$12h(e)\lambda_- = \frac{24h(e)h(D_-)}{\omega(D_-)} \in \mathbb{Z},$$

because $\omega(D_-)$ is 2, 4, or 6. Equations (5.7) and (5.8) therefore give $R_4(D)^3 \in K$ and $R_6(D)^2 \in K$. The seed ratios are nonzero: E_4 vanishes only at the elliptic orbit of discriminant -3 , and E_6 only at that of discriminant -4 , whereas $D < -4$ is fundamental. Adjoining the two positive seed ratios has degree at most $3 \cdot 2 = 6$ over K .

To prove that this one field suffices, use the reduced modular polynomial of (4.12):

$$E_w = E_4^a E_6^b \Delta^m P_w(j), \quad a \in \{0, 1, 2\}, \quad b \in \{0, 1\}, \quad w = 4a + 6b + 12m, \quad (5.20)$$

where $m \geq 0$ and $P_w \in \mathbb{Q}[X]$ is monic of degree m . The algebraic reason is short: the weight condition forces every monomial of $E_w \in \mathbb{Q}[E_4, E_6]$ to have exponents $a+3r, b+2s$ with $r+s = m$. Substituting $E_4^3 = \Delta j$ and $E_6^2 = \Delta(j - 1728)$ gives the displayed form; the constant term $E_w = 1 + O(q)$ makes P_w monic.

The signed absolute product gives the formula without fractional root extraction

$$R_w(D) = R_4(D)^a R_6(D)^b B_{D,\varepsilon}^m \left| \frac{\text{Res}(H_D^+, P_w)}{\text{Res}(H_D^-, P_w)} \right|. \quad (5.21)$$

The resultant ratio is in K , because both genus factors have coefficients there. Its absolute value is also in K , which is a real field. Since the seed forms and Δ never vanish at these CM points, the denominator condition on R_w guarantees a nonzero denominator resultant. This proves the field assertion. $\square$

Corollary 5.5 (Quadratic closure for the recorded discriminants). *For $D = -15, -20, -39$, every defined genus ratio $R_w(D)$ with even $w \geq 4$ lies in $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{5})$, or $\mathbb{Q}(\sqrt{13})$, respectively. Each is computable in that quadratic field using the reduced polynomial P_w , whose degree is at most $\lfloor w/12 \rfloor$.*

Proof. Theorems 5.2 and 5.3 place both seed ratios in the indicated quadratic field. Formula (5.21) completes the proof. $\square$

For example, $E_{14} = E_4^2 E_6$ gives $R_{14}(D) = R_4(D)^2 R_6(D)$. At weight 16, $a = m = 1, b = 0$ and $P_{16}(X) = X - 3456000/3617$, so

$$R_{16}(D) = R_4(D) B_{D,\varepsilon} \left| \frac{H_D^+(3456000/3617)}{H_D^-(3456000/3617)} \right|.$$

The equal degrees of the two genus factors cancel the signs in the linear-polynomial resultants. This formula, together with the three explicit pairs $H_D^\pm$, generates further exact quadratic identities using rational arithmetic alone.

5.5 Certificates and further directions

The companion `cm_genus.py` imports the directed-interval Lambert-series evaluator from `cm_norms.py`. It first certifies the integral Hilbert polynomials, then verifies the displayed genus factorizations by exact symbolic multiplication. For every reduced form it records a represented integer witnessing the genus character, and proves that the opposite genus factor excludes zero at its CM j -value. It next checks the cubic and square identities exactly. The receipt `cm_genus_receipt.json` separates these proof steps from six ordinary 120-digit Fourier-series regressions. A second checker, `verify_genus_arithmetic.py`, independently verifies all six radical equalities and their positive-root choices using only exact standard-library rational arithmetic in the quadratic fields. The numerical regressions have relative discrepancies below 3.1×10^{-120} and are independent consistency checks, not the justification for replacing the manuscript's numerical evidence by theorems.

The genus theorem answers the three recorded ratio questions while leaving several precise extensions open for this project.

1. Extend the genus-ratio certificate to nonmaximal quadratic orders, including the local conductor factors in the twisted Kronecker factorization. The untwisted conductor correction proved earlier cannot simply be copied into a signed genus product.
2. Beyond the three discriminants of Corollary 5.5, classify the discriminants and weights for which the raw ratio $R_w(D)$ itself belongs to $\mathbb{Q}(\sqrt{e})$, rather than merely having a controlled algebraic power in that field. Exact root extraction and representative phases must be part of the statement.
3. For class groups with several independent genus characters, combine their signed formulas by finite Fourier inversion to recover the product over every genus. This determines genus products but does not distinguish classes inside one genus.
4. Investigate twists by nonreal class characters. Their Hecke L -functions no longer reduce to a pair of quadratic Dirichlet L -functions; the resulting period data measure exactly the information absent from genus theory.

6 A self-contained analytic foundation for the zero results

The new zero-count arguments require a bound valid on the entire positive axis, including multiple zeros. We give the necessary proof here. This section reproduces the mechanism already established in the baseline; the subsequent classical and deformation results are the continuation.

For $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, set

$$D_{n,k}^\rho(a) = \sum_{m \geq 0} \rho^m \frac{d^k}{da^k} \frac{\log^n(a+m)}{a+m}. \quad (6.1)$$

The summands are $O(m^{-k-1}(1 + \log^n m))$ on compact positive a intervals. Thus this series and all its fixed a derivatives converge normally, uniformly in $\rho \in [0, 1]$. For $\rho < 1$ it equals the k th derivative of $(-1)^n \partial_s^n \Phi(\rho, s, a)|_{s=1}$.

At $\rho = 1$, differentiating the Hurwitz identity

$$\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a)$$

in a right half-plane and continuing to $s = 1$ proves $D_{n,k}^1 = \gamma_n^{(k)}$. More explicitly, if

$$P_k(z) = \prod_{j=1}^k (1 + z/j) = \sum_i e_i(k) z^i,$$

then coefficient comparison at $s = 1 + z$ gives

$$D_{n,k}^\rho(a) = (-1)^{n+k} k! n! \sum_{i+j=n} \frac{e_i(k)}{j!} \partial_s^j \Phi(\rho, k+1, a). \quad (6.2)$$

At $\rho = 1$, the last Φ denotes ζ . The pole in (1.2) is independent of a , which is why no pole term remains after a positive number of parameter derivatives. This is the harmonic-coordinate version of the all-order formula treated by Coffey [7].

6.1 The positive weight and the reciprocal-gamma polynomial

Define monic Appell polynomials by

$$\frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}. \quad (6.3)$$

For example, with $y = x + \gamma$,

$$Q_1 = y, \quad Q_2 = y^2 - \zeta(2), \quad Q_3 = y^3 - 3\zeta(2)y + 2\zeta(3).$$

The Mellin formula and $(1+z)_k/\Gamma(k+1+z) = 1/\Gamma(1+z)$ give

$$(-1)^{n+k} D_{n,k}^\rho(a) = \int_0^\infty \frac{t^k e^{-at}}{1 - \rho e^{-t}} Q_n(\log t) dt. \quad (6.4)$$

To justify coefficient extraction and differentiation, use

$$1 \leq \frac{1}{1 - \rho e^{-t}} \leq \frac{1}{1 - e^{-t}} \leq 1 + \frac{1}{t}.$$

Near zero the integrable majorants are $t^{k-1-\epsilon}(1 + |\log t|^n)$ with $0 < \epsilon < k$; at infinity a positive lower bound on a supplies exponential decay. Any fixed number of a derivatives only multiplies by a power of t . These estimates also prove continuity at $\rho = 1$.

Lemma 6.1 (Root preservation and strictness). *If a real polynomial p has only real roots and $c \neq 0$ is real, then $p + cp'$ has only real roots. A root of multiplicity $m \geq 2$ is inherited with multiplicity $m - 1$; all other roots are simple. Consequently any sequence of $n - 1$ such operators makes a degree- n real-rooted polynomial strictly real-rooted.*

Proof. Away from the distinct roots r_i of p , of multiplicities m_i , the new roots satisfy

$$\frac{p'}{p} = \sum_i \frac{m_i}{x - r_i} = -\frac{1}{c}.$$

The left side has strictly negative derivative on each complementary interval. Its endpoint limits give one simple solution in each gap and one in the appropriate exterior interval. Factoring at r_i gives the inherited multiplicity, and the degree accounts for all roots. Repeated application reduces every possible multiplicity without introducing new multiple roots. $\square$

Proposition 6.2. *For each $n \geq 1$, Q_n has n distinct real roots. The zeros of Q_{n-1} strictly interlace them.*

Proof. Use the locally uniform Weierstrass product [6]

$$\frac{1}{\Gamma(1+z)} = e^{\gamma z} \prod_{m \geq 1} (1 + z/m) e^{-z/m}.$$

For a finite product, extracting $n![z^n]$ after multiplying by e^{xz} amounts to applying real translations and the operators $1 + \partial_x/m$ to x^n . Lemma 6.1 proves real rootedness of every such polynomial. Coefficient convergence preserves real rootedness for monic polynomials of fixed degree.

Simplicity requires an additional step: a limit of polynomials with simple roots need not have simple roots. Remove the first $n - 1$ factors from the infinite product. The same argument makes the polynomial associated with the tail real-rooted. Restore those $n - 1$ factors; their translations commute with the differential operators, and the lemma makes every root simple. Finally $Q'_n = nQ_{n-1}$ gives strict interlacing by Rolle's theorem. $\square$

6.2 The global bound counts multiplicities

Lemma 6.3 (Laplace variation bound). *Suppose $g(t)$ has exactly N sign changes on $(0, \infty)$, with a fixed nonzero sign between successive sign-change points up to null sets. Suppose its exponentially weighted polynomial moments are absolutely integrable for every positive Laplace parameter. Then $G(a) = \int_0^\infty e^{-at} g(t) dt$ is nonzero as a function and has at most N zeros on $(0, \infty)$, counted with multiplicity.*

Proof. For $N = 0$ the integral has fixed strict sign. For $N > 0$, choose a sign-change point t_0 . The function $(t_0 - t)g(t)$ has one fewer sign change, and its transform is

$$(\partial_a + t_0)G(a) = e^{-t_0 a} \frac{d}{da} (e^{t_0 a} G(a)).$$

Induction bounds its zeros by $N - 1$. If any finite collection of zeros of G has total multiplicity M , Rolle's theorem with multiplicity gives at least $M - 1$ zeros of the displayed derivative, including the inherited endpoint multiplicities. Hence $M \leq N$. This applies to every finite collection. The induction also proves nonvanishing: an identically zero G would have an identically zero transformed derivative. $\square$

Theorem 6.4 (Baseline global zero theorem, reproduced). *For every $n \geq 0$, $k \geq 1$, and $0 \leq \rho \leq 1$, $D_{n,k}^\rho$ has at most n positive zeros, counted with multiplicity.*

Proof. In (6.4), the factor $t^k/(1 - \rho e^{-t})$ is strictly positive. Proposition 6.2 says that $Q_n(\log t)$ has exactly n simple sign changes. The domination already proved supplies the moment hypotheses of Lemma 6.3. That lemma proves the conclusion, including nonvanishing. $\square$

This bound will serve two distinct purposes. At a derivative order with n certified sign changes it proves completeness and simplicity. At a parameter where a double zero is known to exist, it can rule out higher degeneracy once the other zeros use the remaining available multiplicity. The latter use is what makes the Lerch bifurcation proof rigorous.

7 The first classical failure of saturation: the fifth index

The first four Stieltjes indices in the baseline have the maximum possible number of positive parameter-derivative zeros at every derivative order. That pattern fails at the next index. The failure is not restricted to a Lerch deformation: it occurs at the classical Hurwitz endpoint itself. We prove the exact transition

$$\#\{a > 0 : \gamma_5^{(k)}(a) = 0\} = \begin{cases} 3, & k = 1, 2, \\ 5, & k \geq 3, \end{cases} \quad (7.1)$$

and every zero in this display is simple. Thus the least derivative order after which fifth-index saturation persists is exactly three.

Two ingredients distinguish this conclusion from a numerical zero scan. A general upper bound of five controls the entire positive axis at order three. Certified signs at its stationary points then count the zeros at orders two and one by monotonicity. The large positive zeros near 148.9122 and 319.6184 are essential to this argument; a scan confined to a small neighborhood of $a = 1$ misses them.

7.1 Endpoint signs, parity, and monotonicity of zero counts

Retain the Laurent convention

$$\zeta(s, a) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(a) (s-1)^n, \quad a > 0.$$

For the classical family write

$$U_{n,k}(a) = (-1)^{n+k} \gamma_n^{(k)}(a), \quad k \geq 1.$$

Then $U'_{n,k} = -U_{n,k+1}$. The established Appell–Laplace formula is

$$U_{n,k}(a) = \int_0^\infty \frac{t^k e^{-at}}{1 - e^{-t}} Q_n(\log t) dt, \quad \frac{e^{xz}}{\Gamma(1+z)} = \sum_{n \geq 0} Q_n(x) \frac{z^n}{n!}. \quad (7.2)$$

The reciprocal-gamma product makes Q_n strictly real-rooted, and Laplace variation diminution therefore bounds the positive zero count by n , including multiplicities. These are the baseline’s global zero theorem and its proof mechanism, reproduced in Section 6 for completeness. The classical parameter-derivative identity underlying (7.2) is also contained in Coffey’s all-order formulas [7].

For fixed $n \geq 0$ and $k \geq 1$ the endpoint expansions are

$$U_{n,k}(a) \sim k!a^{-k-1} \log^n(1/a), \quad a \downarrow 0, \quad (7.3)$$

$$U_{n,k}(a) \sim (-1)^n(k-1)!a^{-k} \log^n a, \quad a \rightarrow \infty. \quad (7.4)$$

When $n = 0$, the logarithmic powers mean one. The first formula follows either by scaling the integral or by differentiating the shift term $\log^n a/a$ in $\gamma_n(a) = \log^n a/a + \gamma_n(a+1)$. For the second, set $t = u/a$ in (7.2) and use $(1 - e^{-u/a})^{-1} = a/u + O(1 + u/a)$ together with monicity of Q_n . The bound $(1 - e^{-v})^{-1} \leq 1 + v^{-1}$ supplies an integrable majorant after expanding the fixed polynomial in $\log u - \log a$. In particular, every $U_{n,k}$ tends to zero at infinity and has fixed strict signs near both endpoints.

Proposition 7.1 (Parity and persistence before saturation). *Let $N_{n,k}$ be the total multiplicity of the positive zeros of $\gamma_n^{(k)}$. Then*

$$0 \leq N_{n,k} \leq n, \quad N_{n,k} \equiv n \pmod{2}, \quad N_{n,k+1} \geq N_{n,k}. \quad (7.5)$$

Consequently every strict increase in the total zero count is by an even integer. The same assertions hold, at each fixed $0 \leq \rho \leq 1$, for the baseline Stieltjes–Lerch family $D_{n,k}^\rho$.

Proof. The upper bound is the global variation theorem. The endpoint signs in (7.3)–(7.4) show that the total multiplicity has parity n : a real analytic zero reverses sign exactly when its multiplicity is odd.

Suppose the distinct zeros of $U_{n,k}$ are $r_1 < \dots < r_q$, with multiplicities $m_1, \dots, m_q$ and total $N = \sum_i m_i > 0$. Its derivative has inherited multiplicity $m_i - 1$ at each r_i , and Rolle’s theorem supplies a zero between each successive pair. These account for at least $(N - q) + (q - 1) = N - 1$ derivative zeros counted with multiplicity. There is one more stationary point strictly after r_q : a nonzero function leaving zero at r_q , with constant sign afterwards and limit zero at infinity, must attain an interior extremum. Thus $U_{n,k+1} = -U'_{n,k}$ has at least N zeros counted with multiplicity. The case $N = 0$ is immediate.

For fixed $\rho < 1$, the analogous Mellin weight $(1 - \rho e^{-t})^{-1}$ is positive and tends to $(1 - \rho)^{-1}$ at zero. The large- a leading term is then $(-1)^n k! (1 - \rho)^{-1} a^{-k-1} \log^n a$; the small- a leading term is unchanged. The same endpoint and Rolle arguments apply, including the possible multiple zeros at $\rho = 0$. $\square$

Proposition 7.2 (A classical lower bound from matching endpoint data). *For every $k \geq 1$, $\gamma_n^{(k)}$ has at least two distinct positive zeros if $n \geq 2$ is even, and at least three distinct positive zeros if $n \geq 3$ is odd. In particular, saturation for $n = 2, 3$ at every derivative order follows analytically, without a finite sign certificate.*

Proof. For $n \geq 2$, the shift identity at $a = 1$ and its first derivative give

$$\gamma_n(1) = \gamma_n(2), \quad \gamma'_n(1) = \gamma'_n(2), \quad \int_1^2 \gamma'_n(a) da = 0.$$

Put $g = \gamma'_n$. This is a nonzero real analytic function. If its common endpoint value is positive, its integral forces a negative interior value; hence there are at least two sign-changing zeros in $(1, 2)$. The same conclusion holds with signs reversed when the common endpoint value is negative. If the common endpoint value is zero, the integral forces both signs in the interior, and there are at least three distinct zeros in $[1, 2]$, including the endpoints.

For odd n in the nonzero-endpoint case, the two interior sign-changing zeros have odd multiplicities. The odd total parity from Proposition 7.1 forces an additional odd-multiplicity zero, distinct from them. This proves the asserted lower bounds at $k = 1$. Rolle's theorem between distinct zeros, together with the stationary point after the last zero used above, preserves a lower bound on the number of distinct zeros at every subsequent derivative order. The global upper bound makes the two or three zeros simple when $n = 2$ or $n = 3$. $\square$

Thus the count three in (7.1) is the smallest possible classical fifth-index count. Passing from order two to order three raises the positive zero count by exactly two.

7.2 A finite certificate that controls the whole positive axis

For the remainder of the proof abbreviate

$$U_k = U_{5,k}, \quad V_k(a) = \frac{a^{k+1}}{k!} U_k(a).$$

The normalization V_k preserves signs, but its stationary points are not the zeros of U_{k+1} . All monotonicity arguments below are applied to U_k , while V_k is used only for stable sign enclosures.

Let $\varepsilon = 10^{-16}$. Each row of Table 1 specifies a rational interval $[\ell, \ell + \varepsilon]$. The stated endpoint signs of V_k are certified by the outward computation proved in the next subsection.

k	j	ℓ	sign at ℓ	sign at $\ell + 10^{-16}$
3	1	0.9409528529131887	+	−
3	2	1.1292935180347063	−	+
3	3	1.6357795548293846	+	−
3	4	6.1361342793896981	−	+
3	5	319.6184025568026870	+	−
2	1	1.3713298168772550	+	−
2	2	1.9113619446041762	−	+
2	3	148.9122045692035618	+	−
1	1	1.1378900542449414	+	−
1	2	1.6389372768448503	−	+
1	3	2.1249028498579805	+	−

Table 1: Rational isolating intervals of width 10^{-16} . These endpoint signs and the critical-value signs in the next table are finite certificates. The proof establishes uniqueness and completeness.

Denote the five third-order roots by $c_1 < \dots < c_5$ and the three second-order roots by $d_1 < d_2 < d_3$. The interval computation also encloses the preceding-order function uniformly on every indicated root interval. Table 2 lists rational enclosures with outward rounding; they apply in particular at the true roots.

Theorem 7.3 (The exact fifth-index transition). *The positive zero counts are exactly those in (7.1). All zeros are simple. The eleven intervals in Table 1 isolate all zeros at derivative orders one, two, and three. For every $k \geq 3$, the five zeros at order $k + 1$ strictly interlace to the right of those at order k .*

Proof. The five third-order interval signs give at least five distinct zeros of U_3 . The global upper bound of five, counting multiplicity, shows that they are all its positive zeros and that each is simple.

Quantity	Certified rational enclosure
$V_2(c_1)$	[0.006154632340, 0.006154632341]
$V_2(c_2)$	[0.041073722584, 0.041073722585]
$V_2(c_3)$	[-0.097090480335, -0.097090480334]
$V_2(c_4)$	[106.545043234228, 106.545043234229]
$V_2(c_5)$	[-135615.425337743697, -135615.425337743695]
$V_1(d_1)$	[-0.014754703693, -0.014754703692]
$V_1(d_2)$	[0.028710652775, 0.028710652776]
$V_1(d_3)$	[-466915.576951780840, -466915.576951780837]

Table 2: Signs of the lower-order function at every critical point. The verifier encloses the function on the entire rational root bracket, so these signs do not presume a floating-point approximation to a root.

Starting with the positive left endpoint sign, the signs of U_3 on the six complementary intervals are

$$+, -, +, -, +, -.$$

Consequently $U_2' = -U_3$ has constant nonzero sign on each such interval. Its values at the five critical points have signs

$$\text{sgn}(U_2(c_1), \dots, U_2(c_5)) = (+, +, -, +, -).$$

Therefore there is exactly one second-order zero in each of (c_2, c_3) , (c_3, c_4) , and (c_4, c_5) , and no zero in $(0, c_1)$ or (c_1, c_2) . After c_5 , the function U_2 is strictly increasing from a negative value to its limit zero, so there is no zero there either. The three zeros lie away from the critical points, so each is simple. Their brackets in Table 1 consequently identify them as d_1, d_2, d_3 .

The signs of U_2 are now $+, -, +, -$ on the four intervals cut out by the d_j , and $U_1' = -U_2$. The critical-value signs are $-, +, -$. Since U_1 is positive near zero, strict monotonicity gives exactly one zero in each of $(0, d_1)$, (d_1, d_2) , and (d_2, d_3) . After d_3 , U_1 increases from a negative value to zero, so it has no further zero. Again all three crossings are simple.

Finally, five simple zeros at order three force at least five distinct zeros at the next order: Rolle supplies four between adjacent roots and the vanishing tail supplies one after the last root. The global upper bound gives exactly five simple zeros, with no additional zero before the first interval. Repeating proves saturation and strict right interlacing for every $k \geq 3$. $\square$

This proof uses no claim that a dense set of sampled values accounts for every zero. Completeness at the highest checked order comes from a global analytic theorem; completeness at lower orders comes from the complete list of stationary points and their certified signs.

7.3 Rational Euler–Maclaurin verification

The certificate is self-contained in `certify_n5_zeros.py`. It uses only Python’s standard-library integer and rational arithmetic. Its output `n5_zero_certificates.json` records all integer end-points, the analytic error bounds, and the assertions that the relevant intervals exclude zero. The following derivation specifies its mathematical contract.

Define exact rational coefficients by

$$\prod_{j=1}^r (1 + z/j) = \sum_i e_i(r) z^i, \quad R_{n,r}(v) = n! \sum_{i=0}^{\min(n,r)} e_i(r) \frac{(-v)^{n-i}}{(n-i)!}.$$

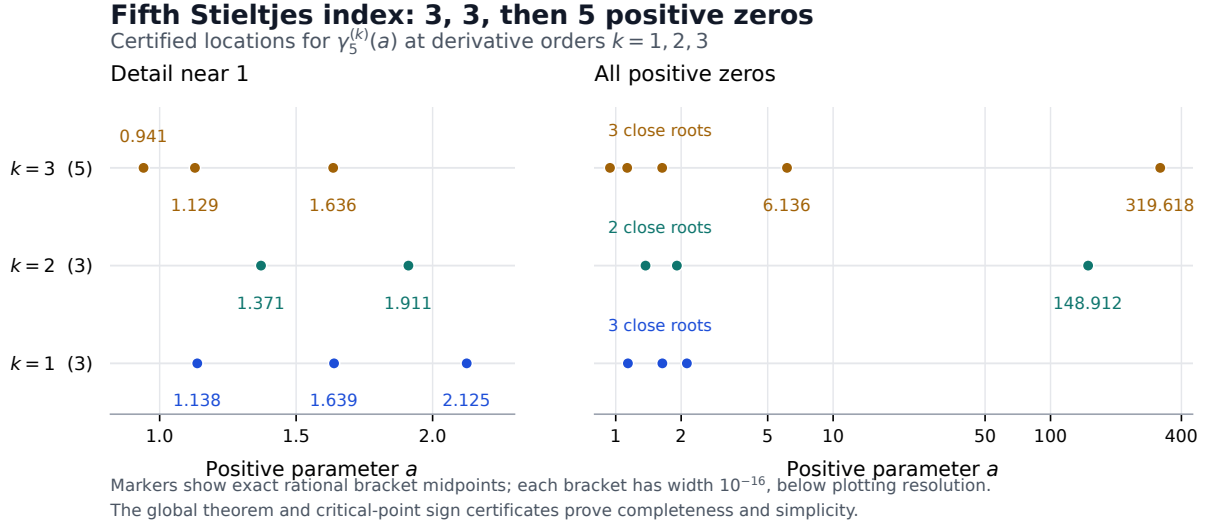


Figure 1: The eleven certified zero locations. The local panel separates the near-unit roots; the logarithmic panel includes the large roots. The brackets have width 10^{-16} , much smaller than a plotted marker.

The spectral representation is

$$V_{n,k}(a) := \frac{a^{k+1}U_{n,k}(a)}{k!} = \sum_{m \geq 0} \left(\frac{a}{a+m} \right)^{k+1} R_{n,k}(\log(a+m)). \quad (7.6)$$

In the three orders used here,

$$\begin{aligned} R_{5,1}(v) &= -v^5 + 5v^4, \\ R_{5,2}(v) &= -v^5 + \frac{15}{2}v^4 - 10v^3, \\ R_{5,3}(v) &= -v^5 + \frac{55}{6}v^4 - 20v^3 + 10v^2. \end{aligned}$$

Higher r in the error analysis use the same exact recurrence, without numerically evaluating zeta or gamma functions.

For positive rational a , choose integers $M, R \geq 1$, put $p = k + 1$, $A = a + M > 1$, $v = \log A$, and $L = k + 2R$. Set

$$\begin{aligned} E_{M,R} &= \sum_{m=0}^{M-1} \left(\frac{a}{a+m} \right)^p R_{n,k}(\log(a+m)) \\ &\quad + \left(\frac{a}{A} \right)^p \left[\frac{A}{k} R_{n,k-1}(v) + \frac{1}{2} R_{n,k}(v) \right] \\ &\quad + \left(\frac{a}{A} \right)^p \sum_{r=1}^R \frac{B_{2r}}{(2r)!} \frac{(k+1)_{2r-1}}{A^{2r-1}} R_{n,k+2r-1}(v), \end{aligned} \quad (7.7)$$

where $(b)_j$ is the rising factorial. Then

$$|V_{n,k}(a) - E_{M,R}| \leq \left(\frac{a}{A}\right)^p \frac{|B_{2R}|}{(2R)!} \frac{(k+1)_{2R-1}}{A^{2R-1}} \mathcal{B}_{n,L}(v), \quad (7.8)$$

$$\mathcal{B}_{n,L}(v) = \sum_{i=0}^{\min(n,L)} \sum_{j=0}^{n-i} \frac{n!e_i(L)}{(n-i-j)!L^j} v^{n-i-j}. \quad (7.9)$$

Every factor in the latter bound is nonnegative.

For clarity, the periodic-Bernoulli error estimate in (7.8) is sufficient; no first-omitted-term sign assertion is needed. With $f(x) = x^{-k-1}R_{n,k}(\log x)$, direct coefficient differentiation gives

$$f^{(q)}(x) = (-1)^q (k+1)_q x^{-k-q-1} R_{n,k+q}(\log x).$$

The integral of f from A to infinity is $A^{-k}R_{n,k-1}(\log A)/k$, which produces the integral term in (7.7). Euler–Maclaurin through the R th Bernoulli correction gives the remaining terms. Its remainder is bounded by $|B_{2R}|/(2R)!$ times $a^{k+1} \int_A^\infty |f^{(2R)}(x)| dx$. Majorize $R_{n,L}$ by its coefficientwise absolute values at $\log x > 0$ and use

$$\int_A^\infty x^{-L-1} \log^d x dx = \frac{A^{-L}}{L} \sum_{j=0}^d \frac{d!}{(d-j)!L^j} \log^{d-j} A.$$

The leading L cancels the last factor of $(k+1)_{2R}$, yielding exactly (7.8)–(7.9).

For logarithms, reduce a positive rational argument to $2^b y$ with $1 \leq y < 2$, set $u = (y-1)/(y+1)$, and use

$$0 \leq \log y - 2 \sum_{j=0}^{T-1} \frac{u^{2j+1}}{2j+1} \leq \frac{2u^{2T+1}}{(2T+1)(1-u^2)}. \quad (7.10)$$

The same formula with $y = 2$ encloses $\log 2$. Exact rational Bernoulli numbers come from their finite triangular recurrence, while the $e_i(r)$ come from multiplying the finite products above.

The executed parameters are

$$n = 5, \quad M = 24, \quad R = 16, \quad T = 120, \quad \text{fixed denominator } 10^{80}.$$

Every addition, multiplication, division, and integer power is rounded outwards at this denominator. The largest analytic Euler–Maclaurin error bound among all thirty enclosures is less than 2.591×10^{-32} . This statement concerns the analytic truncation error; the separate rational interval widths also include all arithmetic rounding and, where appropriate, the uncertainty across a root bracket. The final signs are proved by the complete intervals, not by comparing only the analytic error with a floating-point value.

The code checks twenty-two endpoint signs and eight uniform critical-point signs. Evaluating the same interval formulas with a ranging over a rational interval remains valid by ordinary interval arithmetic and monotonicity of the logarithm. This is how Table 2 certifies values at roots without assigning an exact value to a numerical approximation.

7.4 Consequences and further questions

The baseline correctly limited its all-order small-index theorem to $n \leq 4$. The new fifth-index theorem therefore resolves its first explicitly open small-order case, rather than correcting a theorem already stated with the necessary restriction. It rules out extending the conclusion to all n at every $k \geq 1$.

Several precise continuation questions now replace that false extrapolation. Determine the classical saturation threshold

$$K_n^{\text{cl}} = \min\{k \geq 1 : N_{n,k} = n \text{ and all positive zeros are simple}\}.$$

The present results give $K_n^{\text{cl}} = 1$ for $1 \leq n \leq 4$ and $K_5^{\text{cl}} = 3$. Monotonicity and parity constrain every possible count sequence, while the concentration theorem guarantees an eventual threshold at each fixed index. An effective upper bound in terms of the extreme Appell-root spacing would turn this qualitative fact into a terminating certified search.

The descending critical-value method applies beyond the fifth index: once one derivative order is globally saturated and its roots are isolated, uniform enclosures of the preceding derivative at those critical points determine the complete lower-order zero count. It is particularly useful when the positive roots span several orders of magnitude. The count sequence need not increase at every step; the plateau 3, 3 followed by 5 is already an exact example.

Finally, quantify how the additional classical zeros arise as the Lerch parameter increases from zero to one. The lower bound in Proposition 7.2 uses exact unit-shift endpoint identities at the classical endpoint and does not hold for arbitrary $\rho < 1$. This gives a structural reason to study branch creation as well as the motion of existing simple zeros.

8 Small positive Lerch parameters and real-zero bifurcations

The uniform eventual-zero theorem in the baseline holds with the Laurent index fixed and the derivative order sufficiently large. At a fixed small derivative order, the deformation parameter has a substantially different effect. A multiple real zero at the elementary endpoint can split into nonreal zeros, and real branches that do exist can move in opposite directions. We give an exact classification near that endpoint.

8.1 An elementary polynomial controls the splitting

For integers $n \geq 1$, $k \geq 1$, set

$$h_{n,k}(a) = \frac{d^k}{da^k} \frac{\log^n a}{a}, \quad D_{n,k}^\rho(a) = \sum_{m=0}^{\infty} \rho^m h_{n,k}(a+m) \quad (0 \leq \rho \leq 1). \quad (8.1)$$

At $\rho = 1$ this is $\gamma_n^{(k)}(a)$. The series in (8.1) converges absolutely even there because $k \geq 1$. For $\rho < 1$ it is the k th parameter derivative of $(-1)^n \partial_s^n \Phi(\rho, s, a)|_{s=1}$, so it agrees with the Lerch family used in the manuscript. The underlying Lerch series and Mellin integral are classical [4, 7].

Define integer polynomials $\mathcal{R}_{n,k}$ by

$$\mathcal{R}_{n,0}(x) = x^n, \quad \mathcal{R}_{n,k+1}(x) = \mathcal{R}'_{n,k}(x) - (k+1)\mathcal{R}_{n,k}(x). \quad (8.2)$$

Repeated differentiation gives the exact identity

$$h_{n,k}(a) = a^{-k-1} \mathcal{R}_{n,k}(\log a). \quad (8.3)$$

This recurrence is an efficient way to decide the finite sign appearing in the theorem below; no evaluation of a special function is needed.

Lemma 8.1. *If $n > k$, then $h_{n,k}$ has a zero of multiplicity $r = n - k$ at $a = 1$ and exactly k further positive zeros, all simple and greater than 1. If $n \leq k$, all its n positive zeros are simple and greater than 1. Moreover $h_{n,k}(2) \neq 0$ for all $n, k \geq 1$.*

Proof. Put

$$P_{n,k}(x) = \prod_{j=1}^k (1 + \partial_x/j)x^n.$$

Then $h_{n,k}(a) = (-1)^{n+k}k!a^{-k-1}P_{n,k}(-\log a)$. For a real-rooted polynomial p , the equation $p'/p = -1/c$ shows that $p + cp'$ with $c > 0$ has one simple root in each gap between the distinct roots, one simple exterior root, and only the inherited multiple roots, each with its multiplicity reduced by one. Starting from x^n , after $k < n$ operations the root 0 has multiplicity $n - k$ and all k other roots are simple. Every coefficient of $P_{n,k}$ is nonnegative, and its nonzero coefficients have consecutive degrees. Thus none of its nonzero real roots is positive. After the $k = n$ step the constant coefficient is positive; all roots are then strictly negative and simple, and later steps preserve these properties. This proves the zero assertions after $a = e^{-x}$.

The polynomial $\mathcal{R}_{n,k}$ is nonzero with integer coefficients. If $\mathcal{R}_{n,k}(\log 2)$ vanished, $\log 2$ would be algebraic, contrary to the Hermite–Lindemann theorem [8]. Equivalently, the nontrivial zeros of $h_{n,k}$ are exponentials of nonzero algebraic numbers and cannot equal the algebraic number 2. $\square$

Theorem 8.2 (Complete small-parameter zero count). *Fix $n, k \geq 1$. If $k \geq n - 1$, then, for every sufficiently small $\rho > 0$, $D_{n,k}^\rho$ has exactly n positive zeros, all simple. If $k < n - 1$, write $r = n - k \geq 2$ and $H = h_{n,k}(2) \neq 0$. For every sufficiently small $\rho > 0$, all positive zeros are simple and their number is*

$$N_{n,k}(\rho) = \begin{cases} k + 1, & r \text{ odd}, \\ k + 2, & r \text{ even and } H < 0, \\ k, & r \text{ even and } H > 0. \end{cases} \quad (8.4)$$

More precisely, when $k < n$ put $r = n - k$. Near $a = 1$, all r complex zeros can be written as convergent series in $\varepsilon = \rho^{1/r}$, with leading terms

$$a_\omega(\rho) = 1 + \omega\rho^{1/r} + O(\rho^{2/r}), \quad \omega^r = -\frac{r!}{n!} h_{n,k}(2). \quad (8.5)$$

Every other zero $\alpha > 1$ of $h_{n,k}$ continues analytically as

$$\alpha(\rho) = \alpha - \frac{h_{n,k}(\alpha + 1)}{h'_{n,k}(\alpha)}\rho + O(\rho^2). \quad (8.6)$$

The phrase “sufficiently small” may depend on n, k ; the theorem does not assert a uniform quantitative threshold as those indices grow.

Proof. First suppose $k < n$ and put $r = n - k$. On a complex disk $|a - 1| < \delta < 1$, all logarithms in the series have a consistent principal branch and the series is jointly analytic in (a, ρ) for $|\rho| < 1$. With $a = 1 + z$, Taylor expansion gives

$$D_{n,k}^\rho(1 + z) = \frac{n!}{r!}z^r + O(z^{r+1}) + \rho\{h_{n,k}(2) + O(z)\} + O(\rho^2). \quad (8.7)$$

For $r \geq 1$, substitute $z = \varepsilon u$, $\rho = \varepsilon^r$ and divide by ε^r . The result extends analytically to $\varepsilon = 0$ and equals $(n!/r!)u^r + H$ there. Its r roots are simple by $H \neq 0$. The analytic implicit-function theorem gives r distinct analytic functions of ε , and Rouché’s theorem on a fixed small disk around $a = 1$ shows that these account for all nearby zeros. For positive real ε , a simple real limiting root gives a real branch by conjugation and uniqueness; a nonreal limiting root stays nonreal. Counting the real roots of $(n!/r!)u^r + H$ gives (8.4). When $r = 1$ the same argument gives a single real simple

continuation. When $k \geq n$ there is no multiple root to analyze. The other k (or n) zeros are handled by the ordinary analytic implicit-function theorem, which also gives (8.6).

It remains to exclude positive zeros that escape these neighborhoods. This is a global, not just local, step. On $[1, \infty)$, $h_{n,k}$ and all its fixed derivatives are bounded. Therefore

$$\sup_{a>0} \left| \sum_{m \geq 1} \rho^m h_{n,k}(a+m) \right| \leq M_{n,k} \frac{\rho}{1-\rho}.$$

Near $a = 0$, the unperturbed term has fixed sign and unbounded magnitude, so a sufficiently small fixed neighborhood of zero remains zero-free. Beyond the largest unperturbed zero, every summand has the same strict sign, except for the possible vanishing of the first summand at the endpoint. Hence no zero can escape to infinity. On the remaining compact set away from the chosen zero neighborhoods, $|h_{n,k}|$ has a positive minimum and the displayed uniform bound applies. These three regions exclude every additional positive zero. $\square$

Corollary 8.3 (First derivatives collapse to one or two zeros). *For each fixed odd $n \geq 3$, $D_{n,1}^\rho$ has exactly one positive zero for every sufficiently small $\rho > 0$. For each fixed even $n \geq 2$ it has exactly two. In the even case the smaller zero satisfies*

$$a_-(\rho) = 1 - \left(\frac{(\log 2)^{n-1}(n - \log 2)}{4n} \right)^{1/(n-1)} \rho^{1/(n-1)} + O(\rho^{2/(n-1)}). \quad (8.8)$$

In both cases the large zero tends to e^n .

Proof. Here $h_{n,1}(a) = a^{-2} \log^{n-1} a (n - \log a)$, so $h_{n,1}(2) > 0$ and $r = n - 1$. Apply Theorem 8.2. $\square$

$n \backslash k$	1	2	3	4	5	6	7
2	2	2	2	2	2	2	2
3	1	3	3	3	3	3	3
4	2	2	4	4	4	4	4
5	1	3	3	5	5	5	5
6	2	2	4	6	6	6	6
7	1	3	3	5	7	7	7
8	2	2	4	4	6	6	8

Table 3: The number of positive zeros for sufficiently small positive ρ , with n, k fixed. All signs needed for this table, and the extended table through $n = 16$, are certified using rational bounds on $\log 2$. This table concerns the Lerch endpoint regime, not the classical endpoint $\rho = 1$.

8.2 Alternating motion of odd-index first-derivative zeros

The eventual large- k offset in the baseline decreases with ρ at every fixed zero label. That asymptotic fact does not extend to all small derivative orders. The following exact inequality resolves the issue for odd-index first derivatives.

Theorem 8.4 (Direction of motion at every real zero). *Let $n \geq 1$ be odd, $0 < \rho < 1$, and $D(a, \rho) = D_{n,1}^\rho(a)$. Then every positive zero lies below e^n , and*

$$\partial_\rho D(a, \rho) < 0 \quad \text{whenever } D(a, \rho) = 0. \quad (8.9)$$

For each fixed $a > 0$, there is at most one zero in the parameter $0 < \rho < 1$. On any parameter interval with exactly N positive simple zeros, numbered increasingly, their motion alternates:

$$\frac{da_j}{d\rho} < 0 \quad (j \text{ odd}), \quad \frac{da_j}{d\rho} > 0 \quad (j \text{ even}). \quad (8.10)$$

Here N is necessarily odd.

Proof. Since $n - 1$ is even, $h_{n,1}(t)$ is positive for $0 < t < e^n$, apart from its zero at $t = 1$ when $n > 1$, and negative for $t > e^n$. Thus no zero is possible for $a \geq e^n$ when $\rho > 0$. Put $\lambda = e^n - a$. For every $a > 0$ and $0 < \rho < 1$, termwise differentiation gives the strict inequality

$$\rho \partial_\rho D(a, \rho) - \lambda D(a, \rho) = \sum_{m \geq 0} (m - \lambda) \rho^m h_{n,1}(a + m) < 0. \quad (8.11)$$

Every summand is nonpositive: the factors $m - \lambda$ and $n - \log(a + m)$ have opposite signs. Some summands are strictly negative, and the series converges absolutely. At a zero this proves (8.9). More generally, $\partial_\rho(\rho^{-\lambda} D) < 0$, which proves that there is at most one parameter zero for each fixed a .

The function is positive near $a = 0$ and negative for $a \geq e^n$. Simple real zeros therefore cross with $\partial_a D < 0$ at the odd labels and $\partial_a D > 0$ at the even labels. Implicit differentiation gives $a'_j = -\partial_\rho D / \partial_a D$ and the claimed signs. This argument only asserts differentiability for $\rho < 1$; it makes no endpoint regularity assumption at $\rho = 1$. $\square$

Corollary 8.5 (An increasing middle branch). *For $n = 3$, $k = 1$, there is a number $\rho_0 < 1$ such that for $\rho_0 < \rho < 1$ the three positive simple zeros satisfy*

$$a'_1(\rho) < 0, \quad a'_2(\rho) > 0, \quad a'_3(\rho) < 0.$$

Consequently the proposal that all higher branches decrease with ρ at every derivative order is false. The fixed-index, large-derivative asymptotic statement in the baseline remains valid.

Proof. At $\rho = 1$, the classical $n = 3$ function has exactly three positive simple zeros; an analytic proof is given in Section 7. Uniform convergence of (8.1) and its a derivative on compact positive intervals as $\rho \uparrow 1$ preserves opposite endpoint signs in three disjoint isolating intervals. The global bound of three zeros counted with multiplicity makes these the only zeros and makes them simple. Apply Theorem 8.4. $\square$

8.3 A nondegenerate creation of two real zeros

The following endpoint signs, consistent with the baseline small-index certificates, are verified afresh by the rational checker in this package:

$$D_{3,1}^1(4/5) > 0, \quad D_{3,1}^1(11/10) < 0, \quad D_{3,1}^1(3/2) > 0.$$

These three signs are also recomputed in the present package. They suffice to make the bifurcation statement fully analytic.

Theorem 8.6 (A real saddle-node in the Lerch deformation). *There are $0 < \rho_* < 1$ and $4/5 < a_* < 3/2$ such that*

$$D_{3,1}^{\rho_*}(a_*) = \partial_a D_{3,1}^{\rho_*}(a_*) = 0, \quad \partial_a^2 D_{3,1}^{\rho_*}(a_*) > 0, \quad \partial_\rho D_{3,1}^{\rho_*}(a_*) < 0.$$

This is the first parameter at which a zero occurs in $[4/5, 3/2]$; there is only one double zero there at that parameter. For every $\rho_* < \rho < 1$, the interval contains exactly two simple zeros and there is exactly one further positive zero. Near ρ_* the two zeros have the expansions

$$a_{\pm}(\rho) = a_* \pm \sqrt{-\frac{2\partial_{\rho}D(a_*, \rho_*)}{\partial_a^2 D(a_*, \rho_*)}} \sqrt{\rho - \rho_*} + O(\rho - \rho_*). \quad (8.12)$$

The smaller branch decreases and the middle branch increases throughout $(\rho_*, 1)$.

Proof. For a fixed a , (8.11) shows that a negative value, once reached as ρ increases, remains negative. Since the two endpoint values are positive at $\rho = 1$, they are positive for every $0 < \rho < 1$. Corollary 8.3 and its proof show that D is strictly positive on the whole interval $[4/5, 3/2]$ for all sufficiently small positive ρ . The negative value at $11/10$ for $\rho = 1$ persists for parameters sufficiently close to one. Compactness and continuity therefore give a first parameter $\rho_* \in (0, 1)$ where the minimum on the interval is zero. More explicitly, take

$$\rho_* = \inf\{0 < \rho < 1 : \min_{a \in [4/5, 3/2]} D(a, \rho) \leq 0\}.$$

The restriction $\rho > 0$ excludes the multiple zero of the elementary endpoint itself. The preceding small-parameter positivity keeps this infimum away from zero. That minimum is attained at an interior point a_* , where $D = \partial_a D = 0$.

For every $0 < \rho < 1$, positivity at $3/2$ and negativity at e^3 force a further sign-changing zero in $(3/2, e^3)$. At ρ_* , the double-or-higher zero in $[4/5, 3/2]$ together with that further zero uses at least three units of multiplicity. The global bound of three therefore makes the first zero exactly double, makes it unique in the first interval, and excludes every additional zero. Since it is a local minimum, $\partial_a^2 D > 0$. Equation (8.9) gives $\partial_{\rho} D < 0$.

For every $\rho > \rho_*$, the value at the fixed point a_* is negative. The positive endpoints give two distinct zeros in the interval, and the right-hand sign change gives a third. The global bound makes all three simple and excludes others. Their directions follow from Theorem 8.4. Finally the analytic Taylor expansion at the double zero, or the implicit-function theorem applied with $\sqrt{\rho - \rho_*}$ as local parameter, gives (8.12). $\square$

The supplied quadrature diagnostic locates the fold and illustrates the square-root law. Those decimal estimates are not used to prove the bifurcation, its nondegeneracy, or the direction of the middle branch. The theorem does not claim that no other changes in the rightmost zero configuration occur below ρ_* ; a complete classification of that earlier parameter interval is a separate question.

9 Corrections, conjectures, and further research

9.1 Two precise corrections to the pinned text

A consequence is not an equivalence. In `05-signed-kernels.tex`, immediately after `signed:eq:rank-conjecture`, the manuscript calls the pair of rank formulas equivalent to a statement about the dimension of the Gaussian-supported row space. If V_w is the row space, projection onto the non-Gaussian columns gives exactly

$$\dim(V_w \cap \mathbb{Q}^{\{g\text{-columns}\}}) = \text{rank } A_w - \text{rank } A_w^{-g}.$$

Thus the two individual ranks imply the stated dimension, while their difference alone does not determine either individual rank. Replace “Equivalently” by “Consequently,” and replace the

following sentence about the equivalence by a sentence about this dimension formula. Theorem 2.1 now proves both ranks, so the conjecture can also be promoted to a theorem when its proof is integrated. The companion patch makes the narrow logical correction without assuming a particular chapter-insertion scheme.

Even modified polylogarithms on real arguments. In 03-algebraic.tex, equation `bloch:eq:Pm` defines

$$P_m(x) = \operatorname{Re}_m \left(\sum_{j=0}^{m-1} \frac{2^j B_j}{j!} \log^j |x| \operatorname{Li}_{m-j}(x) \right), \quad \operatorname{Re}_m = \begin{cases} \Re, & m \text{ odd}, \\ \Im, & m \text{ even}. \end{cases}$$

The later paragraph on higher golden ladders says that the same coordinates turn a real weight-six ladder into a $\zeta(6)$ multiple. With this definition, P_6 is identically zero on real arguments. The text itself correctly states this parity fact a little later. In particular, for the golden arguments in $(0, 1)$ every summand inside the imaginary part is visibly real. A nonzero multiple of $\zeta(6)$ cannot be the value of that real-argument modified-polylogarithm combination.

The correction is local: remove the weight-six assertion in these coordinates, retain the raw classical-polylogarithm identity at whatever status its independent evidence supports, and check odd-weight conversions directly from the definition. The supplied replacement text avoids asserting unverified conversions at the other higher weights. No claim that the raw weight-six ladder is false follows from this observation.

9.2 Statements that require qualification, not retraction

The fifth-index theorem does not contradict the baseline theorem, whose small-index conclusion is restricted to $n \leq 4$. It disproves the unrestricted extrapolation of that conclusion. Similarly, the increasing middle Lerch branch does not contradict a large-derivative asymptotic offset formula. It shows that its monotonic direction cannot be imposed on every branch at every small derivative order.

The CM product theorem is a norm statement unless a phase is supplied. The real-part convention at discriminant -15 is mathematically material: the factor $7/8$ cannot be omitted. The absolute norm alone does not license a raw signed product at arbitrary discriminants or weights.

The weight-five witness proves an obstruction only for the specified linear product matrix. No period-independence claim, no impossibility of using higher-depth relations, and no negative answer to the S_4 conjecture follows. This scope was already correctly emphasized by the baseline; the shorter integral witness makes it easier to check.

9.3 A concrete next zero-count conjecture

Conjecture 9.1 (Sixth- and seventh-index first derivatives). *The function γ_6' has exactly four positive zeros and γ_7' has exactly five positive zeros. In each case every positive zero is simple.*

The numerical candidates are

$$\begin{aligned} n = 6 : & \quad 0.8426787120, 1.2904536569, 1.7994089556, 2.2385946523; \\ n = 7 : & \quad 0.7781418150, 0.9499243123, 1.4420456198, \\ & \quad 1.9499308927, 2.3354109288. \end{aligned}$$

These statements are *conjectures*, not further certified theorems. The supplied diagnostic uses 70 decimal digits, an Euler–Maclaurin evaluation with $M = 24$, $R = 16$, and 992 test points at each index: 401 logarithmic points spanning 10^{-3} to $10^{3.1}$ and 591 linear points $j/200$, $60 \leq j \leq 650$. Root refinement produces the displayed values. At first derivative order the spectral summands are negative once $a > e^n$, so this upper tail is analytically free of zeros. Nevertheless, the finite scan does not certify the unsampled intervals or exclude a tangency. A proof should use global saturation at a higher derivative order and descend through every critical point, as in Theorem 7.3.

The conjecture is deliberately finite. These two observations do not support a simple universal formula for the first-derivative count: the oscillation near $a = 1$, the wide range of large roots at higher derivative orders, and the count plateaus already proved at index five make extrapolation hazardous.

9.4 A focused program for the formal relation system

Construct certificates without elimination. The proof of Theorem 2.1 identifies a unimodular coordinate system in which the problem breaks into one- and two-variable blocks. Turn that proof into an explicit symbolic certificate compiler. For a target vector, it should either return an integer row combination (or, in even weight, one for twice the target) or return an integral annihilating witness. The witness basis is already explicit, so membership testing can be performed with only $w - 1$ pairings in odd weight. Tracking the affine right sides through the same transformations is the remaining engineering and symbolic-algebra task.

Identify the missing relation for S_4 . The six-entry witness pairs to 7 with the target. Any proposed enlargement must include a relation whose coefficient vector does not annihilate that witness. This provides an exact preliminary test before computing a much larger matrix. Candidates include distribution, argument transformations, and elimination through higher depth. The research question is not merely whether an enlarged matrix has larger rank, but which analytically justified relation kills this particular obstruction and with what right side.

Determine the effect of distribution integrally. The 2-torsion in even weight is a property of the stated product presentation. Does adjoining the appropriate distribution rows remove some or all of it? What is the Smith form of that enlarged lattice? This is a concrete question about an explicitly constructible sequence of integer modules; no unproved transcendence assumption is required.

Generalize the involution method to other levels. At level four, conjugation reduces the colors to six and the residual action becomes the swap $(u, v) \mapsto (v, u)$. At other small levels, determine whether an integral color decomposition produces similarly manageable finite-group actions on homogeneous polynomials. Rational rank formulas alone would miss torsion introduced by nonunimodular changes of basis, so an integral formulation should be maintained from the outset.

9.5 A focused program for CM identities

Automate the exact arithmetic factor. For fixed even weight, generate F_w in exact rational arithmetic and evaluate its resultant with a certified Hilbert class polynomial. At weights four and six the arithmetic is concentrated in $H_D(0)$ and $H_D(1728)$; in higher weights the zeros of the Eisenstein series create additional algebraic factors. A systematic collection of these factors would distinguish patterns forced by modular algebra from patterns inferred from decimal recognition.

Classify phases and genus refinements. Extend the phase analysis along the unit arc to every even weight, including the additional arc zeros of E_w . Extend the proved character products to nonmaximal orders and combine several independent genus characters to recover every genus product. For higher genus rank the plus and minus fibers of one character are unions of genera. These are the natural intermediate objects between a full class norm and an individual CM evaluation. Every claimed formula should specify the reduced representative, automorphy factor, and chosen algebraic embedding.

Move beyond full norms. For larger class groups, determine individual algebraic factors relative to a fixed CM period. For the genus ratios, determine when the uniform degree bound of Theorem 5.4 is attained, and which proper subfields occur as the weight changes. A rational class norm does not imply that raw sums of positive-weight CM values are rational or Galois-invariant. Replacing such raw sums by explicitly normalized modular quantities is the natural route to well-posed algebraicity questions.

9.6 A focused program for zero geometry

Determine saturation thresholds. Set

$$K_n^{\text{cl}} = \min\{k \geq 1 : \gamma_n^{(k)} \text{ has } n \text{ positive simple zeros}\}.$$

The exact values are now $K_n^{\text{cl}} = 1$ for $1 \leq n \leq 4$ and $K_5^{\text{cl}} = 3$. The baseline concentration theorem implies that the threshold exists for every fixed index. Find a useful effective bound in n , and then determine whether the sequence of thresholds has any monotonicity or regular asymptotic behavior. The proved parity and count monotonicity restrict possible answers but do not determine them.

Make the small-parameter theorem effective. Theorem 8.2 gives a complete count for sufficiently small positive ρ , but not an explicit threshold. Its proof suggests a constructive route: isolate the algebraic roots of the integer polynomial, bound derivatives on disjoint disks, and bound the remaining series uniformly. Quantifying the separation from $\log 2$ is the sensitive step. Transcendence proves nonvanishing, while a useful numerical threshold needs an effective separation bound.

Certify the fold coordinates. The existence and nondegeneracy of the index-three fold are proved. The diagnostic predicts

$$a_* \approx 1.04975305088025546, \quad \rho_* \approx 0.999990466837738634.$$

Produce a rational interval box for this pair by applying an interval Newton method to $(D, \partial_a D)$, with explicit quadrature or Euler–Maclaurin remainders in both variables. The determinant of the Jacobian is $-\partial_\rho D \partial_a^2 D > 0$ at the fold, so the proved nondegeneracy identifies the right certification problem. The decimals above are not yet such a box.

Describe the entire deformation, not only its endpoints. For odd n at first derivative order, $\partial_\rho(\rho^{-(e^n-a)} D) < 0$. Thus each fixed a changes sign at most once. Determine the topology of the resulting sign domains: which negative intervals are created, which can merge, and which bifurcations are possible under the global multiplicity bound. For $n = 3$, the proved fold controls all parameters after its first appearance in $[4/5, 3/2]$, while the possible earlier configuration on the right remains a separate question.

Treat the simultaneous large-index regime. The baseline concentration expansion fixes n and lets k grow. The extreme roots of Q_n change with n , and the small-parameter Puiseux exponent $1/(n - k)$ exhibits another nonuniform limit. Determine which double-scaling regimes preserve the known zero approximations, and where the right asymptotic description must change. A satisfactory answer would connect the formal spectral polynomials, reciprocal-gamma root geometry, and the observed threshold sequence.

A Reproducibility, evidence boundaries, and integration

A.1 What the package contains

The archive contains the article source, its component sections, the compiled PDF, vector figures with plotting sources, exact verification programs, numerical diagnostics, and machine-readable receipts. A provenance manifest records the pinned source commit and the SHA-256 digests of the consulted manuscript files. A separate claim ledger maps results to their proofs, their scripts, and their status. The two narrow text corrections are provided as an ordinary unified patch, together with integration notes keyed to existing manuscript labels.

The article is a self-contained continuation, not a silent replacement of the 228-page source manuscript. Each contribution is isolated in a section file so that a repository maintainer can move it to the relevant chapter. Numbered labels have distinct prefixes for the relation-lattice, mixed-family, CM, and classical-zero contributions. The common zero preliminaries can be omitted when integrating into the existing zero chapter, provided the references are redirected to its established theorems.

A.2 Exact computations and analytic certificates

Program	Verified object	Logical status
<code>verify_rank.py</code>	Integer rows, rational ranks, primitive kernel witnesses, and Smith forms in the specified finite range.	Exact algebra; checks the general proof.
<code>verify_mixed_family.py</code>	A finite integer row certificate for twice the target at even weights; independent integral evaluation.	Exact rows; quadrature is diagnostic.
<code>cm_norms.py</code>	Reduced forms, interval-enclosed singular moduli, unique integer class-polynomial coefficients, exact factorizations.	Certified computation using CM integrality.
<code>cm_genus.py</code>	Quadratic-field factors, certified genus labels, and exact cubic and square identities for the ratios.	Exact and interval certificate.
<code>modular_weight_polynomials.py</code>	Exact F_w and coefficient identities through the weight- $12w$ valence bound, $4 \leq w \leq 24$.	Exact modular-form certificate.
<code>certify_n5_zeros.py</code>	Rational interval signs at all eleven root brackets and all eight needed critical values; three index-three endpoint signs.	Finite analytic certificate.
<code>certify_lerch_splitting.py</code>	120 signs of integer polynomials at $\log 2$, for $2 \leq n \leq 16$.	Exact rational enclosures.
<code>lerch_fold_diagnostic.py</code>	A high-precision approximate solution of $D = \partial_a D = 0$.	Numerical diagnostic only.
<code>explore_n6_n7.py</code>	Finite scans and root refinements supporting Conjecture 9.1.	Numerical evidence only.

The zero certificate uses only Python’s integer and rational arithmetic; its logarithm bounds and Euler–Maclaurin remainder are proved in the text. It does not call a numerical Stieltjes-constant routine. The matrix certificate uses exact SymPy arithmetic and an independently constructed matrix. The mixed-family certificate works directly with product rows, and the all-weight proof supplies the uniform row combination.

The six radical genus evaluations also have an independent checker, `verify_genus_arithmetic.py`, which performs arithmetic as rational pairs in the two quadratic fields and checks the positive-root choices. It imports neither SymPy nor the CM verifier.

The CM coefficient certificate uses the directed interval implementation in `mpmath.iv`. Its proof contract is explicit: the elementary interval functions enclose their true values, the rational tail bound is added before division or multiplication, and each resulting coefficient interval has a unique possible integer. Classical CM integrality supplies the fact that the true coefficient is an integer. This is a conventional computer-assisted proof, with the interval implementation as part of its trusted computational base. No claim of proof-assistant formalization is made.

A.3 Replay and interpretation

From the extracted archive, the principal commands are

```
python3 -m pip install -r requirements.txt
python3 code/verify_all.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The replay driver runs each principal verifier in a fresh output directory, preserving the delivered receipts for comparison. Numerical diagnostics can be run separately with the documented flag. They are excluded from the proof-only interpretation even when they agree to the full working precision.

For the rank replay, the delivered run covers weights 3 through 13 and both Smith forms through weight 12. It checks the 92×23 weight-five matrix and the exact pairing 7 of the short witness with the target. For the mixed family, the finite row certificate is checked at all even weights 4 through 20, and the independent integral at weights 2, 4, 6, 8, 10. The general claims are proved in the article; these finite ranges are validation ranges, not the asserted scope of the theorems.

The article’s PDF was compiled with all cross-references resolved and rendered for visual inspection. Figure labels, tables, long formulas, and page breaks were checked. The archive includes a checksum manifest so that later editorial changes can be distinguished from the delivered research version.

A.4 Suggested integration order

First apply the two local corrections. Next insert the relation-lattice theorem after the formal product system and replace the old odd-weight rank conjecture by a reference to the theorem. Place the mixed-family identity next to the existing Gaussian and signed-color evaluations. In the CM chapter, upgrade the product identities only after inserting the period normalization, certified class data, and phase conventions. In the zero chapter, put the fifth-index transition after the small-index certificates, and the small-parameter and bifurcation results after the definition of the Lerch family. Retain the diagnostic labels on the fold coordinates and the sixth- and seventh-index conjectures.

This integration order preserves the distinction between a theorem, an exactly checkable finite lemma, an independently verified numerical identity, and an open conjecture throughout the combined manuscript.

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